Quantum Games, Walks and Algorithms

A thesis Submitted
in Partial Fulfilment of the Requirements
for the Degree of

MASTER OF SCIENCE

by

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to the

School of Physical Sciences

National Institute of Science Education and Research

Bhubaneswar

30th **April 2016**

DECLARATION

I hereby declare that I am the sole author of this thesis in partial fulfillment of the requirements for a postgraduate degree from National Institute of Science Education and Research (NISER). I authorize NISER to lend this thesis to other institutions or individuals for the purpose of scholarly research.

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School:

Date:

List of publications:

- Namit Anand and Colin Benjamin. "Do quantum strategies always win?." Quantum Information Processing 14.11 (2015): 4027-4038.

ACKNOWLEDGEMENTS

I wish to thank Dr. Colin Benjamin for his patience, support and efforts.

ABSTRACT

Quantum games, walks and algorithms are at the heart of the field of Quantum information and computation. Throughout this thesis I have studied each of these foundational topics in detail and at the same time strived to bridge the gaps between them. We have formally introduced new ideas like a classical strategy beating a quantum strategy, the formal merger of quantum games and quantum walks, and using Leggett- Garg inequalities for quantum state tomography and for quantum to classical transition in quantum games.

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Chapter 1

Introduction

Game theory and physics at first do not seem to have much in common. Games like playing cards and Russian roulette seem to be rather dependent on chance and guesses, rather than some deterministic laws of nature. And yet, it was shown by von Neumann that the consciousness of a player is not an essential part of a game[1]. From an abstract point of view, game theory is about competing interaction between agents/players, fighting to maximize or minimize some numbers namely their profit. What seems more unlikely is for game theory to have anything to do at all with quantum mechanics. However as shown by Meyer, it is interesting to consider what would happen if linear combinations/superpositions of strategies were allowed, i.e. a quantum generalization of game theory is attempted.

Some good reasons why quantum games might be interesting are:

1. Classical game theory has found innumerable applications to economy, psychology, ecology and biology. Since the mathematical formalism of game theory is largely based on probability, there is a natural interest in what happens when this is generalized over quantum probabilities.

2. There are speculations that games of survival are being played at the molecular level, for e.g. "Selfish Genes", an arena where quantum mechanics becomes important.

3. There is a deep connection between game theory and the theory of communication. This isomorphism can be seen as, every time a player passes a decision to the other player,

[1] J. von Neumann and O. Morgenstern, The Theory of Games and Economic Behaviour (Princeton University Press, Princeton, NJ, 1947).

it is a form of communicating information. Since the true laws of physics are quantum, it is important that we think of this information also as quantum information.

Some applications of quantum game theory in the field of Quantum information and communication are:

1. One can design zero-sum games like PQ penny flip by Meyer, where quantum strategies have a distinct advantage.
2. Remote gambling protocols can be analyzed as a quantum game and fairness can be ensured.
3. Mean field games have helped to explain socio-economic data.
4. The modelling of an eavesdropper across a quantum communication channel manifests itself as a quantum game.

Chapter 2

Classical and Quantum Game theory

Game theory is a branch of Mathematics which deals with situations in which decision makers(players) interact.

When one encounters the theory of games, some naturally arising questions are:

What is a game ?

A game is a form of **competitive** activity or sport played according to **rules**.

Why should we study game theory ?

Simply because it is there in nature and like all phenomena in nature, it too is very interesting.

Natural Selection is one of the oldest games played by different species of organisms on earth.

A broad overview of the large arena to which game theory applies goes as follows:

- Economic firms competing for profit.
- Rational decision making in say court verdicts.
- Voting in political scenarios.
- cumulative evolution of organisms together, be it human or animal.

Figure 2.1: Chess - a common strategic game.

2.1 Basic Definitions

Some fundamental definitions involved in the mathematical theory of games are the following:

2.1.1 Theory of Rational Choice

The theory of rational choice is basically states that every player in a game plays to maximize his/her profit, and that when faced upon with a group of choices to pick, will go for the one which has the maximum incentive. This assumption is a commonplace in many theories of decision making since it does not give preference to the individuals likes and dislikes, rather it helps to construct a game independent of individual predilections and biases.

2.1.2 Actions

Every player has a set of possible moves/actions available to him/her for all initial conditions or situations in the game. Let us call the set of all possible actions, A. Along

with this set of actions comes a set of payoffs corresponding to each element in the set of actions. That is, for every move that a player makes, there is a number attached, a number which tells the corresponding profit or loss associated with the move. Given the theory of rational choice and the one-to-one correspondence between the set of actions and the set of payoffs, it means that the player always chooses the move with the higher payoff.

2.1.3 Preferences and payoff functions

As the theory of rational choice dictates, when faced upon with multiple choices, a player chooses the one with the highest payoff. And when the payoffs are equal, the player is indifferent to the choices.

The payoff function is like a functional on the space of actions, it takes an element of the space of actions and maps it to a real number, namely the payoff.

Mathematically if the payoff function is f, then for two actions x,y $\in$ the set A,

$f_x > f_y$ iff the player prefers x to y.

2.2 Strategic Theory

A **STRATEGIC GAME** is a natural model of mutually interacting players/decision-makers. Along with every player, comes naturally the set of possible moves/actions. It is also important at this stage to define some rules of the game since the players must be made aware of what is allowed and what is not. This theory can essentially capture the conflicts and interactions between all the players. This happens since each player is affected by the choices of all the other players and not just their own.

2.2.1 Defintion

A strategic game has the following components:

- a set of players
- for each player, a set of actions
- for each player, and each action, a set of preferences over the set of actions.

One often translates the game into matrix form, or gives the payoff functions representing the players actions and preferences.
An important thing to notice here is that there is no notion of time in such a theory. Although this may seem weird, the idea is that every player chooses their action simultaneously and this is the reason what a strategic game is called a "simultaneous game".

2.2.2 Example: Prisoner's Dilemma

Two thieves plan to rob a bank. As they enter the perimeter, the police arrests them for trespassing. The cops suspect that the thieves had planned to rob, but the lack the evidence to support such a conviction. So the police officer puts them in separate cells and is looking for a confession from them so as to charge them with the robbery.
Having studied game theory in college, the officer uses Prisoner's dilemma on them. He goes to the separate cells and makes the following statement:
You will be charged with trespassing, which amounts to a one month jail sentence. I know you were planning on robbing the bank, but I cannot prove it right now. To do so, I need your testimony. If you cooperate, I will dismiss your trespassing charges, and your partner will be charged to the fullest extent of the law i.e. a twelve month jail sentence. I am offering the same deal to your partner. But if you both confess, then your individual testimony is no longer valuable and so both of you will be charged with eight months of

jail sentence each.

Let's assume both the criminals only care about themselves, and about minimizing their jail time, should they take the officers deal?

This situation can be modeled as a strategic game:

Players: The two suspects.

Actions: Each player's set of actions is Keep Quiet, Confess.

Solving the game

The game can be solved in several ways but one of the simplest ways to condense all that we know, is to put it in a matrix form, also known as the **normal form** representation of a game:

The first entry is the payoff of Player 2 and the second is that of Player 1.

	Keep Quiet	Confess
Keep Quiet	-1,-1	-12,0
Confess	0,-12	-8,-8

To find out the optimal strategy, we consider the action of Player 1(say) arbitrary and look at Player 2's payoff.

Case 1:

Player 1 chooses KEEP QUIET then

	Keep Quiet
Keep Quiet	-1,-1
Confess	0,-12

One can clearly see that Confess is a better option for Player 2 since $0 > -1$

Case 1:

Player 1 chooses CONFESS then

	Confess
Keep Quiet	-12,0
Confess	-8,-8

One can clearly see that Confess is still a better option for Player 2 since $-8 > -12$

Since the game is symmetric, similar is the outcome of an analysis made on the Player 2.

2.2.3 Battle of the Sexes

A husband and wife wish to go out together. They decided on Ballet and Football but forgot they had finalized which one. One person prefers Ballet and the other prefers Football. If they go to different places, each of them is equally unhappy not being able to be together.

The normal form of the game is:

	Ballet	Football
Ballet	2,1	0,0
Football	0,0	1,2

At this point it is very important to first develop the concept of a Nash equilibrium to continue our search for optimal strategy.

2.3 Nash Equilibrium

What actions will be chosen by the players in a strategic game? We wish to assume, as in the theory of a rational decision-maker, that each player chooses the best available action. In a game, the best action for any given player depends, in general, on the other players actions. So when choosing an action a player must have in mind the actions the other players will choose. That is, she must form a belief about the other players actions.

A **Nash equilibrium** is an action profile a^* with the property that no player i can do better by choosing an action different from a_i^* , given that every other player j adheres to a_j^* .

We can state the definition of a Nash equilibrium precisely:
Let a be an action profile, in which the action of each player i is a_i.
Let a_{-i} be any action of player i (either equal to a_i, or different from it). Then ($a_i^{'}$, a_i) denotes the action profile in which every player j except i chooses her action a_j as specified by a, whereas player i chooses $a_i^{'}$. (The -i subscript on a stands for except i.) That is, ($a_i^{'}$, a_{-i}) is the action profile in which all the players other than i adhere to a while i deviates to $a_i^{'}$. (If $a_i^{'} = a_i$ then of course ($a_i^{'}$, a_{-i}) = (a_i, a_{-i}) = a). If there are three players, for example, then ($a_2^{'}$, a_{-2}) is the action profile in which players 1 and 3 adhere to a (player 1 chooses a_1, player 3 chooses a_3) and player 2 deviates to $a_2^{'}$.

2.3.1 Definition:

The action profile a^* in a strategic game is a Nash equilibrium if, for every player i and every action a_i of player i, a^* is at least as good according to player is preferences as the action profile (a_i , a_i^*) in which player i chooses a_i while every other player j chooses a_j^* .

Equivalently, for every player i,

$$u_i(a^*) \geq u_i(a_i, a^*_{-i}) \tag{2.1}$$

for every action a_i of player i, where u_i is a payoff function that represents player is preferences.

2.3.2 Examples

- In Prisoner's Dilemma, the Nash equilibrium is the action profile (Confess,Confess). In particular, the incentive to free ride (with 0 payoff, the maximal in the game)eliminates the possibility that the mutually desirable outcome (Keep Quiet,Keep Quiet) occurs.

- In Battle of the Sexes:
  Consider the action profiles (Ballet,Ballet) , (Ballet,Football) , (Football,Ballet) , (Football,Football).

  - (Ballet, Ballet): If player 1 switches to Football then her payoff decreases from 2 to 0; if player 2 switches to Football then her payoff decreases from 1 to 0. Thus a deviation by either player decreases her payoff. Thus (Ballet, Ballet) is a Nash equilibrium.
  - (Ballet, Football): If player 1 switches to Football then her payoff increases from 0 to 1. Thus (Ballet, Football) is not a Nash equilibrium. (Player 2 can increase her payoff by deviating, too, but to show the pair is not a Nash equilibrium it suffices to show that one player can increase her payoff by deviating.)
  - (Football, Ballet): If player 1 switches to Bach then her payoff increases from 0 to 2. Thus (Football, Ballet) is not a Nash equilibrium.

  - (Football, Football): If player 1 switches to Ballet then her payoff decreases from 1 to 0; if player 2 switches to Bach then her payoff decreases from 2 to 0. Thus a deviation by either player decreases her payoff. Thus (Football, Football) is a Nash equilibrium.

- In Matching Pennies, there is no Nash equilibrium.

2.4 Mixed Strategy Nash Equilibrium

Here is a very simple game. You and I each have a penny. Simultaneously, we choose whether to put our penny on the table with heads or tails facing up. If both of the pennies are heads or both of the pennies are tails (that is, they match), then you have to pay me a dollar. But if one is heads and the other is tails (that is, they do not match), then I have to pay you a dollar.
The matrix form of the game is:

	Heads	Tails
Heads	1,-1	-1,1
Tails	-1,1	1,-1

Matching pennies is an example of a strictly competitive or zero sum game which has no Nash equilibrium.
Does that mean this game has no solution? Not nearly. As it turns out, every finite game has at least one Nash equilibrium. (John Forbes Nash, of A Beautiful Mind fame, proved this result, which is why we call this Nashs theorem. It is also why we search for **Nash** equilibria.) We define a game as being finite if the number of players is finite and the number of pure strategies each player has is finite. Here, there are two players, and each player has two pure strategies. Therefore, matching pennies is a finite game, so it must

have a Nash equilibrium. We know matching pennies has no pure strategy Nash equilibria, so it must have an equilibrium in mixed strategies.

Suppose that player 2 chooses each of her actions with probability $\frac{1}{2}$

- If player 1 chooses Head with probability p and Tail with probability 1 - p then each outcome (Head, Head) and (Head, Tail) occurs with probability $\frac{1}{2}$ p, and each outcome (Tail, Head) and (Tail, Tail) occurs with probability $\frac{1}{2}$(1 - p). Thus player 1 gains 1 with probability $\frac{1}{2}$p $+$ $\frac{1}{2}$(1 - p), which is equal to $\frac{1}{2}$, and loses 1 with probability $\frac{1}{2}$.
- In particular, the probability distribution over outcomes is independent of p! Thus every value of p is optimal. In particular, player 1 can do no better than choose Head with probability $\frac{1}{2}$ and Tail with probability $\frac{1}{2}$.
- A similar analysis shows that player 2 optimally chooses each action with probability $\frac{1}{2}$ when player 1 does so. We conclude that the game has a stochastic steady state in which each player chooses each action with probability $\frac{1}{2}$.

We refer to a probability distribution over strategies as a mixed strategy and strategies with certainty as pure strategy.

2.4.1 Definition

The mixed strategy profile α^* in a strategic game is a (mixed strategy) Nash equilibrium if, for each player i and every mixed strategy α_i of player i, the expected payoff to player i of α^* is at least as large as the expected payoff to player i of (α_i , α^*_{-i}) according

to a payoff function whose expected value represents player is preferences over lotteries. Equivalently, for each player i,

$$U_i(\alpha^*) \geq U_i(\alpha_i, \alpha^*_{-i}) \tag{2.2}$$

for every mixed strategy α_i of player i, where $U_i(\alpha)$ is player is expected payoff to the mixed strategy profile α.

2.4.2 Expected Payoff

A players expected payoff to the mixed strategy profile α is a weighted average of her expected payoffs to all mixed strategy profiles of the type (a_i, α_{-i}), where the weight attached to (a_i, α_{-i}) is the probability $\alpha_i(a_i)$ assigned to a_i by player is mixed strategy α_i.

$$U_i(\alpha) = \sum_{a_i \in A_i} \alpha_i(a_i) U_i(a_i, \alpha_i), \tag{2.3}$$

where Ai is player is set of actions (pure strategies) and $U_i(a_i, \alpha_{-i})$ is her expected payoff when she uses the pure strategy that assigns probability 1 to a_i and every other player j uses her mixed strategy α_j.

2.5 Quantum Games

Quantum game theory is an extension of classical game theory to the quantum domain. It differs from classical game theory in three primary ways:

- Superposed initial states

  The information transfer that occurs during a game can be viewed as a physical process. In the simplest case of a classical game between two players with two strategies each, both the players can use a bit (a '0' or a '1') to convey their choice of strategy. A popular example of such a game is the Prisoners' Dilemma, where each of the convicts can either cooperate or defect: withholding knowledge or revealing that the other committed the crime. In the quantum version of the game, the bit is replaced by the qubit, which is a quantum superposition of two or more base states. In the case of a two-strategy game this can be physically implemented by the use of an entity like the electron which has a superposed spin state, with the base states being $+1/2$ (plus half) and $1/2$ (minus half). Each of the spin states can be used to represent each of the two strategies available to the players. When a measurement is made on the electron, it collapses to one of the base states, thus conveying the strategy used by the player.

- Quantum entanglement of initial states

  The set of qubits which are initially provided to each of the players (to be used to convey their choice of strategy) may be entangled. For instance, an entangled pair of qubits implies that an operation performed on one of the qubits, affects the other qubit as well, thus altering the expected pay-offs of the game.

- Superposition of strategies to be used on the initial states

  The job of a player in a game is to choose a strategy. In terms of bits this means that

the player has to choose between 'flipping' the bit to its opposite state or leaving its current state untouched. When extended to the quantum domain this implies that the player can rotate the qubit to a new state, thus changing the probability amplitudes of each of the base states. Such operations on the qubits are required to be unitary transformations on the initial state of the qubit. This is different from the classical procedure which chooses the strategies with some statistical probabilities.

2.6 Quantizing classical games: The Classical Penny flip game

PQ Penny flip game as discussed by Meyer in his paper[1] "Quantum strategies" is a altered version of the traditional classical game "Matching pennies" which has the following rules:

2.6.1 Matching pennies

- Players P and Q each have a penny and are separated from each other so they can't communicate with each other, however there is an unbiased Referee with whom they communicate simultaneously and disclose the state of their pennies.
- Initial state of the pennies is either of (H,H), (H,T), (T,H) or (T,T).
- Each player can choose to either flip or not flip his/her own penny, independent of each other.
- If at the end of the game both the pennies are heads or both are tails, then P wins else Q wins.

Thus, the payoff matrix of the game is:

$$\begin{array}{cc} & Q \\ & \begin{array}{cc} Head & Tail \end{array} \\ P \begin{array}{c} Head \\ Tails \end{array} & \begin{bmatrix} (1,-1) & (-1,1) \\ (-1,1) & (1,-1) \end{bmatrix} \end{array} \tag{2.4}$$

The numbers in the matrix above are the payoffs for either player, first index is for P and second is for Q. For example $(-1,1)$ means P loses a penny while Q gains a penny as the end state is Heads. This is a zero sum game and there is no pure strategy Nash equilibria here. However there is a mixed strategy Nash equilibria[10]. In the mixed strategy the players repeatedly play the game and as has been shown before in Ref.[10] P and Q both getting exactly 50 % of the times Heads and Tails is the mixed strategy Nash equilibrium at which the payoff of both players is zero.

2.6.2 PQ Penny flip

The PQ penny flip was designed by Meyer in Ref.[1], its a close cousin of the Matching pennies game as described above and has the following rules:

- Players P and Q each have access to a single penny.
- Initial state of the penny is heads(say).
- Each player can choose to either flip or not flip the penny.
- Players cannot see the current state of the penny.
- Sequence of actions : Q $\longrightarrow$ P $\longrightarrow$ Q
- If final state is heads, Q wins else P wins

Since players cannot see the current state of the penny, their actions become independent of each other and so each strategy, flipping or not flipping is equally desirable.

The matrix form of the game is:

$$\begin{array}{cc} & Q \\ & \begin{array}{cccc} NN & NF & FN & FF \end{array} \\ P \begin{array}{c} N \\ F \end{array} & \begin{bmatrix} (-1,1) & (1,-1) & (1,-1) & (-1,1) \\ (1,-1) & (-1,1) & (-1,1) & (1,-1) \end{bmatrix} \end{array} \tag{2.5}$$

where N is not flipping and F is flipping. Both players have $\frac{1}{2}$ as their winning probability. The numbers in the matrix above are the payoffs for either player, first index is for P and second is for Q. For example $(-1,1)$ means P loses a penny while Q gains a penny as end state is Heads. PQ penny flip is an example of a strictly competitive or zero sum game which again has no pure strategy Nash equilibrium but has as before a mixed strategy Nash equilibrium. The pair of mixed strategies of P flipping the penny with probability $\frac{1}{2}$ and Q playing each of the available four strategies with probability $\frac{1}{4}$ is the mixed strategy Nash equilibrium with payoff again zero.

2.7 Quantum Penny Flip game

The Quantum penny flip game as described by Meyer in Ref.[4] is as follows. The starship Enterprise is facing a calamity. This is when Q appears on the bridge and offers to rescue the ship if Captain P can beat him at a simple game: Q produces a penny and asks the captain to place it in a small box, head up. Then Q, followed by P, followed by Q, dip their fingers into the box, without looking at the penny, and either flip it over or leave it as it is. After Q's second turn they open the box and Q wins if the penny is head up. Q wins every time they play, using the following "quantum" strategy:

$$\begin{array}{ccc} |0\rangle & \xmapsto{\text{Q does } H} & \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ & \xmapsto{\text{P does } X \text{ or } I} & \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ & \xmapsto{\text{Q does } H} & |0\rangle \end{array}$$

Here 0 denotes 'head' and 1 denotes 'tail', $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the Hadamard transformation , I means leaving the penny alone and the action with $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ flips the penny over. Q's quantum strategy of putting the penny into the equal superposition of 'head' and 'tail', on his first turn, means that whether Picard flips the penny over or not, it remains in an equal superposition which Q can rotate back to 'head' by applying H again since $H = H^{-1}$. So Q always wins when they open the box. Thus playing the penny flip game with a quantum strategy enables the player to win against one who is playing classical, 100% of the time. Of course one may ask what if both play quantum? In that case as shown by Meyer, the quantum advantage vanishes, see Theorem 2 of Ref.[4]:A two-person zero-sum game need not have a quantum/quantum equilibrium.

2.8 A nonlocal game: The GHZ Game

I present the following game for a team of three players.

- The players are allowed to make any preparations before they are taken to three remote locations A,B and C.
- Then, at a certain time t, each player is asked one of two possible questions: "What is X?" or "What is Y ?".
- Each player must give an answer which is also limited to only two possibilities: "1" or "-1".
- They have to give their answer quickly, i.e., before the time they might receive a message sent by another player after the time t.
- According to the rules of the game, either all players are asked the X question, or only one player is asked the X question and the other two are asked the Y question.

- The team wins if the product of their three answers is -1 in the case of three X questions and is 1 in the case of one X and two Y questions.
- My friend is asked what should the team do in order to win for sure.

2.8.1 Analysing the game

Since each player is not able to get any message from the other players about which questions they were asked before the time she has to give an answer, it seems that she cannot gain anything by delaying the decision of which answer to give for each question until the question is actually asked.

Thus, an optimal strategy should correspond to prior definite decisions of each player which answers to give for possible questions. But it is easy to prove that any such strategy cannot ensure winning for all allowed combinations of questions.

Indeed, if it does, then there must be a set of answers $X_A, Y_A, X_B, Y_B, X_C, Y_C$, where X_A is the answer of the player in A on question X, etc., such that the following equations are fulfilled.

$$X_A X_B X_C = -1, \tag{2.6}$$

$$X_A Y_B Y_C = 1, \tag{2.7}$$

$$Y_A X_B Y_C = 1, \tag{2.8}$$

$$Y_A Y_B X_C = 1. \tag{2.9}$$

This however is impossible, because the product of all left hand sides of equations is the product of squares of numbers which are $+1$ or -1 and therefore it equals 1, while the product of all right hand sides of these equation yields 1.

The particles are prepared in a entangled state (which is usually called the GHZ state):

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|1_z\rangle_A|1_z\rangle_B|1_z\rangle_C - |0_z\rangle_A|0_z\rangle_B|0_z\rangle_C). \tag{2.10}$$

Now, if a member of the team is asked the X question, she measures σ_x and gives the answer which she obtains in this experiment. If she is asked the Y question, she measures σ_y instead. Quantum theory ensures that the team following this strategy always wins.

The states satisfy the above games of the rule:

$$\sigma_{A_x}\sigma_{B_x}\sigma_{C_x} = -1 \tag{2.11}$$

$$\sigma_{A_x}\sigma_{B_y}\sigma_{C_y} = 1 \tag{2.12}$$

$$\sigma_{A_y}\sigma_{B_x}\sigma_{C_y} = 1 \tag{2.13}$$

$$\sigma_{A_y}\sigma_{B_y}\sigma_{C_x} = 1 \tag{2.14}$$

Let us show, for example, that the first equation is true. In the spin x bases for all particles the GHZ state (which is given in the spin z bases) is:
$|GHZ\rangle = \frac{1}{4}[(|1_x\rangle_A + |0_x\rangle_A)(|1_x\rangle_B + |0_x\rangle_B)(|1_x\rangle_C + |0_x\rangle_C) - (|1_x\rangle_A - |0_x\rangle_A)(|1_x\rangle_B - |0_x\rangle_B)(|1_x\rangle_C - |0_x\rangle_C)]$
which becomes
$|GHZ\rangle = \frac{1}{2}[|1_x\rangle_A|1_x\rangle_B|0_x\rangle_C + |1_x\rangle_A|0_x\rangle_B|1_x\rangle_C) + |0_x\rangle_A|1_x\rangle_B|1_x\rangle_C + |1_x\rangle_A|0_x\rangle_B|0_x\rangle_C]$
Therefore, we see explicitly that the GHZ state is a superposition of states for each of which $\sigma_{A_x}\sigma_{B_x}\sigma_{C_x} = -1$.

In the quantum solution of the problem the players do not decide in advance the answers they'll give for each out of the two possible questions. In the "proof" of the impossibility of this task presented above it was erroneously assumed that delaying the decision which

answer to give until the time the question is asked cannot help. The assumption sounds plausible since relativistic causality prevented sending signals after the time the questions were asked, but, nevertheless, the assumption is wrong because it does not take into account unusual correlations which quantum objects can exhibit.

2.8.2 Why GHZ game is popular ?

Performing the game with spin$\frac{1}{2}$ particles as described above might serve as an experimental proof of the nonexistence of local hidden variables which produce agreement with experiment. Suppose, however, that Nature is governed by local hidden variables. This means that the results σ_{A_x}, σ_{A_y}, σ_{B_x}, σ_{B_y}, σ_{C_x}, σ_{C_y} exist prior to their measurements and, therefore, there are answers $X_A, Y_A, X_B, Y_B, X_C, Y_C$ prior to the time the questions are asked. These answers can fulfill at most three out of four equations (1) and, therefore, in the game in which the four question patterns are chosen with equal probability, on average there will be at most a 75 per cent success rate of any team playing this game. If a team shows on average a higher result, this is an experimental proof that Nature is not governed by local hidden variables.

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Chapter 3

Can classical strategies win against quantum strategies?

[1]

3.1 Introduction

Game theory is an extremely interesting and sophisticated field which holds within its ambit the power to resolve conflicts, propose new strategies in making war and peace and understand various situations wherein competing entities interact. Examples range from zero sum games like tick-tack-toe to non zero sum games like Prisoners dilemma. However, in the past 15 years a new addition to game theory has come into being. This is the story of Quantum games. It studies how to quantize the previously played classical games or propose new games which explicitly use quantum mechanical phenomena like entanglement, nonlocality and quantum interference. Further, in contradistinction to the just mentioned definition of quantum games, sometimes a player may use quantum mechanics as a viable strategy to defeat his/her opponent in a classical game which is also included in the general framework of quantum games. In the context of "quantized games", Meyer was one of the first to detect and prove that Quantum strategies if used by a player in the classical Penny flip game can help a player outwit her opponent 100% to nil. This led to a flourishing field of quantum games. Previous works include quantum prisoners dilemma [2], including an experimental implementation of the Prisoner's dilemma in a NMR quantum computer[3]. Recently, comparisons have been made be-

[1] Anand, Namit, and Colin Benjamin. "Do quantum strategies always win?." Quantum Information Processing 14.11 (2015): 4027-4038.

tween classical correlations and quantum correlations(entanglement) in quantum games with some interesting conclusions-like correlated noise can reproduce features of quantum games with entanglement[5].

One of the key motivations for playing games in the quantum world, comes from the possibility of re-formulating quantum communication protocols, and algorithms, in terms of games between quantum and classical players[6]. In fact Ref.[6] claims that there exists a basic relationship between quantum algorithms and quantum games and research on quantum games can lead to new quantum algorithms[7].

However there have been works which have commented that quantum games do not offer anything new and whatever they promise can be replicated via classical correlated equilibrium[4]. In this work we apply this hypothesis to the entanglement based quantum penny flip game. We see that indeed a classical player can outwit her quantum opponent by using a particular mixed strategy which is not possible in the non-entanglement based quantum penny flip game. Further, Ref. [8], questions two aspects of quantum game theory- first, whether the quantum strategy solves the underlying classical problem and to what extent the new solution can be obtained via a classical model. Ref. [8] gives the example of Shor's algorithm, in the language of game theory, this would be called a quantum strategy which solves the classical factoring problem (the game). Ref. [8] also questions that the quantum prisoners dilemma doesn't really solve the classical prisoners dilemma. In the entangled quantum penny flip game of ours we are looking at a new game of preserving versus destroying a maximally entangled state in which counter intuitively a classical mixed strategy can defeat the hitherto unbeatable quantum strategy of Meyer's penny flip game[1]. In other words, making an analogy to Shor's algorithm (quantum strategy) which solves the classical factoring problem (classical game), our work shows that a classical mixed strategy (classical algorithm) can beat the quantum strategy (quantum algorithm) in winning (solving) the entangled penny flip game (quantum problem).

Stretching this analogy further, this therefore could in fact be of relevance to the field of quantum algorithms whose justification stems from the fact that they are indeed more efficient than classical algorithms. We in this work put forth a counter example which demonstrates that a particular classical algorithm can outwit the previously unbeatable quantum algorithm in the entangled quantum penny flip problem.

In recent years a move to distinguish between the application of quantum mechanics to game theory known as "quantized games" and the introduction of game theory to quantum mechanics known as "gaming the quantum" has also been made. Reference [12] delineates this distinction in proper detail. "Quantized games" refers to the fact that effective quantization of a game occurs when it collapses to the underlying classical game on introduction of restrictions, similar to the general notion of quantization. However, in the absence of any restrictions one is gaming the quantum[12]. So if a gamed quantum system does not reduce to the underlying classical game on application of certain restrictions then it is an example of gaming the quantum while one which reduces to the underlying classical game on imposing these restrictions is an example of a quantized game. In this work we show how our "entangled penny flip game" under certain restrictions(no entanglement and no superposition) collapses to the classical matching pennies game, thus is an example of "gaming the quantum" rather than of "quantized game".

The entangled penny flip game could be mistaken for the two penny flip game[9]. In the latter there are two pennies to begin with, however in the former there is a Bell state to begin with, which is the original penny of the classical penny flip. The two states of the penny, heads and tails are now the maximally entangled or the completely separable states. There is however a crucial difference between the classical penny flip and the entangled quantum penny flip. In the former there is no 'draw' while in the latter the final state could be in an non-maximally entangled state, in which case the game ends in a draw. Our game like PQ penny flip is also played sequentially. Only difference being P

and Q only work on the part of the entangled state in their possession.

3.2 Quantum entangled penny flip game: Introduction

We begin with a maximally entangled state of two qubits which is shared by P and Q and allow P and Q to make moves on only the qubit in their possession. If the final state of the game is a maximally entangled state then Q wins, and if it is a non-maximally entangled state then its a draw, if its a separable state then P wins. We allow P to make classical moves, i.e. either I or X and Q to make Hadamard transforms (or any unitary operations) on his qubit. This game should not be confused with the quantum two penny game[9] wherein there are two heads and two tails, here the heads of the penny is the "entangled state" and tails is the "separable state". It is the entanglement analog of the classical penny flip game with the addition of a "draw".

3.2.1 Quantum entangled penny flip game: The classical pure strategy

In this case the classical player P is allowed only the pure strategy of either flipping or not flipping his qubits. Consider the initial state of the system as the entangled state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle) \tag{3.1}$$

Sequence of actions is : Q $\longrightarrow$ P $\longrightarrow$ Q.

Step 1:

So, if Q does a Hadamard transformation on his qubit we get:

$$H \otimes I(\frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)) = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle) \tag{3.2}$$

So the current state of the game is:

$$|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle) \tag{3.3}$$

Step 2:

Now its P's turn to make a move, which means either leaving the state unchanged or applying the flip X operation, when applying the flip the state becomes-

$$I \otimes X(\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)) = \frac{1}{2}(|01\rangle - |00\rangle - |11\rangle - |10\rangle) \tag{3.4}$$

So the current state of the game is either of:

$$|\psi\rangle = \frac{1}{2}(|01\rangle - |00\rangle - |11\rangle - |10\rangle) \tag{3.5}$$

or

$$|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle) \tag{3.6}$$

Step 3:

Q's final move on the above states leaves us with:

$$H \otimes I(\frac{1}{\sqrt{2}}\,|01\rangle - |00\rangle - |11\rangle - |10\rangle) = |B1\rangle = \frac{1}{\sqrt{2}}(|11\rangle - |00\rangle) \tag{3.7}$$

or

$$H \otimes I(\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{3.8}$$

So finally we have the either the following states, depending on whether P had flipped or not flipped his qubit:

- $|B\rangle = \frac{1}{\sqrt{2}}(|11\rangle - |00\rangle)$
- $|B1\rangle = \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle)$

Matching pennies with entangled states and one player having quantum strategies while the other is classical, like the original version by Meyer gives a definite win to Q as both the states above are maximally entangled Bell states. The game here is about whether player Q having all quantum strategies at his hand can keep the state maximally

entangled, whereas P with classical moves can or cannot reduce the entanglement. If in the end all the states obtained are maximally entangled then Q wins, if they are separable then P wins. What the game essentially shows is that it is not possible for player P with pure classical strategies to effect the quantum correlations between the particles in a way which Q cannot revive with her quantum moves.

3.2.2 Quantum entangled penny flip game : The classical mixed strategy

Now we allow for P using albeit classical but mixed strategy. This entails P with probability p flipping the state of his qubit and with probability $1-p$ leaving it as it is. In this case as before P and Q start with a maximally entangled Bell pair. Q makes the first move, again a Hadamard. Next is P's turn and as defined earlier flips with probability p. Finally Q does a Hadamard. At the end the final state is observed for the amount of entanglement. As before P and Q share a Bell pair. But contrary to the previous case, P now uses a mixed classical strategy, Q still uses the pure quantum strategy. Since P uses a mixed strategy we have to take recourse to density matrices to explain the results.

Step 1: The initial state

In the form of density matrices the initial state $|\psi\rangle = \frac{1}{\sqrt{2}}[|10\rangle - |01\rangle]$ is represented as-

$$\rho_0 = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Step 2: Q makes her move

Q makes an unitary transformation on her part of the shared state. $U_{Q_1} = \begin{bmatrix} a & b^* \\ b & -a^* \end{bmatrix}$. The state after Q's move then is $\rho_1 = (U_{Q1} \otimes I)\rho_0(U_{Q1} \otimes I)^\dagger$.

Step 3: P plays mixed

P as we said earlier in contrast to the previous section plays a mixed strategy, which

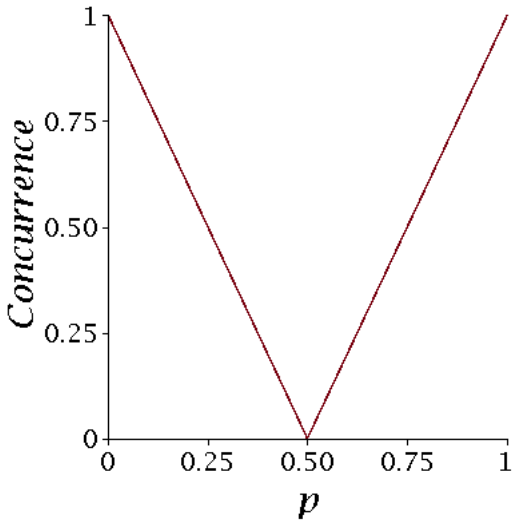

Figure 3.1: Concurrence vs p showing that entanglement vanishes at $p = 1/2$, so by P's classical moves entanglement is completely destroyed enabling him to win.

entails flipping the state of his qubit with probability "p" or not flipping. The state after P's move then is: $\rho_2 = p(I \otimes X)\rho_1(I \otimes X)^\dagger + (1-p)(I \otimes I)\rho_1(I \otimes I)^\dagger$.

Step 4: Q makes her final move

At the end Q makes her final move, which as before has to be an unitary transformation, it further could be same as her first move or different. Thus $U_{Q2} = \begin{bmatrix} \alpha & \beta^* \\ \beta & -\alpha^* \end{bmatrix}$. The state after this final move then is $\rho_3 = (U_{Q2} \otimes I)\rho_2(U_{Q2} \otimes I)^\dagger$.

To understand this case of P using mixed, lets analyze this case for Q using the familiar Hadamard transform in both steps 2 and 4. In this special case,

$$\rho_3 = \frac{1}{2}\begin{bmatrix} p & 0 & 0 & -p \\ 0 & 1-p & -1+p & 0 \\ 0 & -1+p & 1-p & 0 \\ -p & 0 & 0 & p \end{bmatrix}.$$

To check the entanglement content of this final state we take recourse to an entanglement measure- Concurrence . Concurrence for a two qubit density matrix ρ_3 is defined as follows- we first define a "spin-flipped" density matrix, γ as $(\sigma_y \otimes \sigma_y)\rho_3^*(\sigma_y \otimes \sigma_y)$. Then

we calculate the square root of the eigenvalues of the matrix $\rho_3\gamma$ (say λ_1,λ_2,λ_3,λ_4) in decreasing order. Then, Concurrence is :

$$\text{max } (\lambda_1 - \lambda_2 - \lambda_3 - \lambda_4, 0)$$

In Fig. 1, the Concurrence is plotted. At $p = 1/2$ concurrence vanishes implying a separable state and a win for P's mixed classical strategy. Thus proving our contention that there exists a classical mixed strategy which can defeat a quantum strategy.

Quantum entangled penny flip game: General quantum strategy versus mixed classical strategy

In the above two cases we had the Quantum player playing the identical quantum strategy of applying a Hadamard at his turn. Now what if he uses a general unitary not restricted to just a Hadamard. Further in successive turns he does not implement the same unitary, i.e., $U_{Q_1} \neq U_{Q_2}$. We again see that as in the previous case when confronted with P playing a mixed classical strategy the quantum player is beaten for the case when $p = 1/2$. We implement general quantum strategy by just two changes to the scheme introduced in the above sub-section. Replacing the Hadamard in Step 2 with U_{Q_1} and the Hadamard in Step 4 with U_{Q_2}. $U_{Q_i} = \begin{bmatrix} \cos(\theta_i)e^{i\phi_i} & \sin(\theta_i)e^{i\phi'_i} \\ \sin(\theta_i)e^{-i\phi'_i} & -\cos(\theta_i)e^{-i\phi_i} \end{bmatrix}, i = 1, 2.$

In Fig. 2 we plot the Concurrence when the quantum player does not play with just Hadamard but does a general unitary transformation to the qubit in his possession. We only plot the Concurrence versus θ_1 and θ_2, to show that the classical player wins for $p = 1/2$. One can similarly plot the Concurrence versus $\phi_1, \phi_2, \phi'_1, \phi'_2$ and get identical results to what has been plotted in Fig. 2, so we do not repeat them here.

3.3 Quantum circuit implementation

To design the quantum circuit for this game, we first have a Referee/Arbiter who generates the Bell state $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ as shown in Fig. 3. He then hands over this state to P and

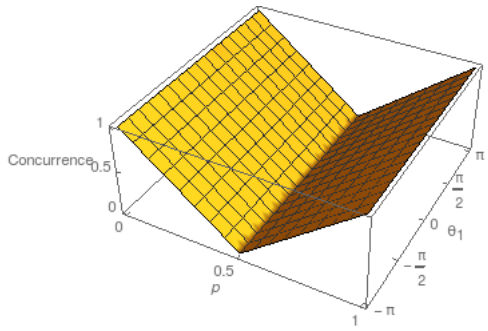

Figure 3.2: Concurrence vs. θ_1, $\theta_2 = 0, \phi_1 = \pi/2, \phi_1' = 0, \phi_2 = \pi/2, \phi_2' = 0$

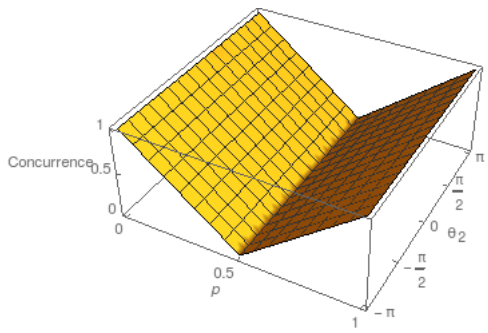

Figure 3.3: Concurrence vs. θ_2, $\theta_1 = 0, \phi_1 = \pi/2, \phi_1' = 0, \phi_2 = \pi/2, \phi_2' = 0$

Figure 3.4: The Concurrence when quantum player plays a general unitary vs. classical players mixed strategy. The classical player always wins when $p = 1/2$, confirming that regardless of whether quantum player uses a Hadamard or any other unitary he always loses when classical player plays a mixed strategy of either flipping or not flipping with probability 50%.

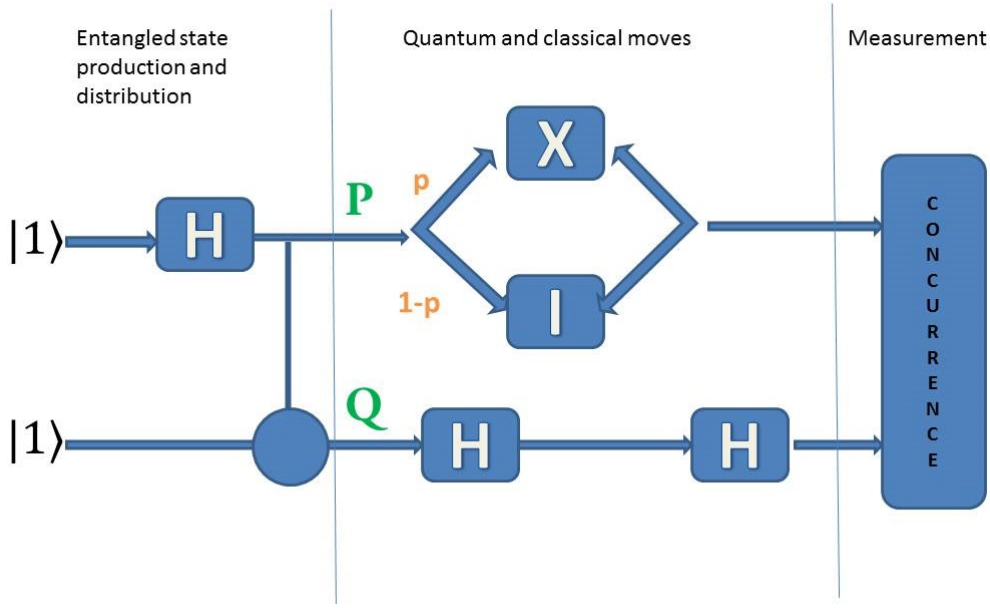

Figure 3.5: Quantum player uses Hadamard

Q, who now share this maximally entangled 2-qubit state. P and Q get access to their respective qubits from the Referee. The operations they undertake on their respective qubits proceeds as $Q \rightarrow P \rightarrow Q$. Q first does a unitary (Hadamard) on her qubit and then P with probability p flips his qubit or with probability 1-p leaves it unaltered. Finally, Q does another unitary (Hadamard, again). The Referee now comes back into the picture where he takes the final output state and establishes the entanglement content via the Concurrence. The Concurrence of a general two qubit mixed state has been experimentally determined in Ref.[11]. If the Referee finds that the Concurrence vanishes (Separable state) then P is declared winner, on the other hand if the Concurrence is one (Maximal entanglement), then Q wins, in the rest of the cases the honors are equally shared i.e. it's a draw.

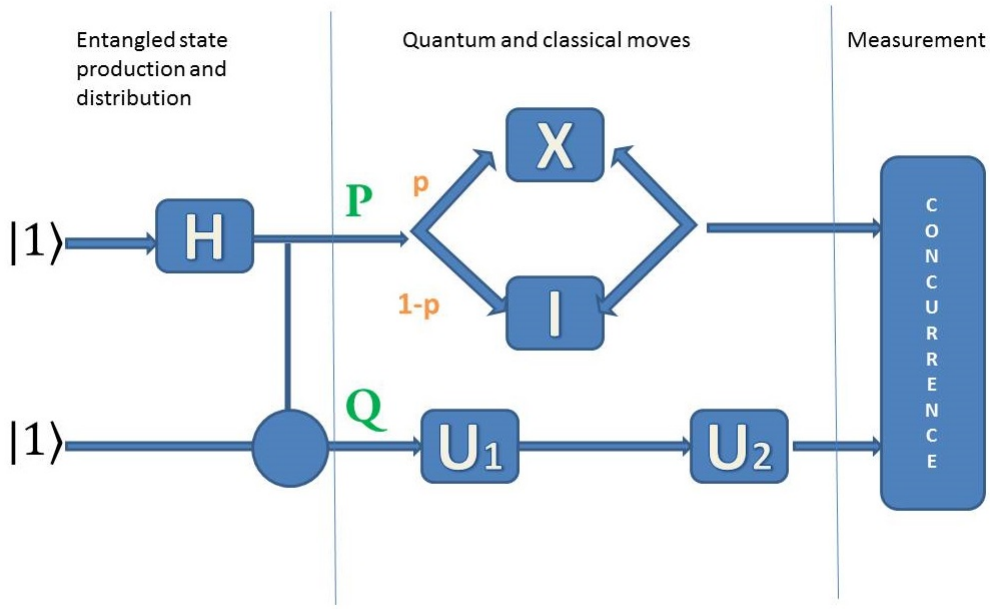

Figure 3.6: Quantum player uses a general unitary

Figure 3.7: The quantum circuit for the entangled penny flip game. M denotes measurement of entanglement content via concurrence, see Ref.[11].

3.4 Gaming the quantum vs quantized games: Finding the underlying classical game

In Ref. [12] the authors delineate the ideas in the theory of quantum games into introducing elements of game theory in quantum mechanics(gaming the quantum) vs using quantum mechanics to study game theory (quantized games). Our entangled quantum penny flip game is an example of gaming the quantum, since when restrictions are put "no entanglement" and "no superposition" the game reduces to the classical matching pennies game.

Meyer in his paper Quantum strategies uses the PQ penny flip game closely related to the traditional Matching pennies game. The following restrictions transform our Quantum entangled penny flip game to the original Matching pennies game. If we destroy the entanglement between the qubits and allow both the players to make only classical moves (Q has no recourse now to a Hadamard and has to satisfy herself with a flip or no-flip), the results show the same outcomes as the matching pennies game as shown below:

We begin with the referee destroying the entangled state $|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)$ shared between P and Q: The state of the system is now either $|\psi\rangle = |10\rangle$ or $|01\rangle$. The sequence of actions by either player is same as before: Q $\longrightarrow$ P $\longrightarrow$ Q, which puts the state of the system in either of the following four product states: $|\psi\rangle = |00\rangle$ or $|01\rangle$ or $|10\rangle$ or $|11\rangle$.

If the state is $|\psi\rangle = |00\rangle$ or $|11\rangle$, i.e., both bits in the same state then Q wins. and if the state is $|\psi\rangle = |01\rangle$ or $|10\rangle$ i.e., both bit states don't match then P wins. In this way our Quantum entangled penny flip game reduces to the underlying Matching pennies game. Hence satisfying the criteria for a "quantized game" as described in ref [12].

3.5 Conclusions

The Quantum penny flip game with entangled particles has nontrivial outcomes as compared to the original quantum penny flip game[1]. In a particular case where classical player uses a mixed strategy with p = "0.5", the quantum player indeed loses as opposed to the expected win for all possible unitaries!

The first takeaway from this work with particular relevance to game theory is that some one using a classical strategy can beat someone with a quantum strategy. Meyer showed that in the PQ penny flip if both players use Quantum strategies then there is no advantage. However, a player using a quantum strategy will win 100% of the time against a player using a classical strategy because of an enlarged strategy space due to quantum superpositions. We wanted to check the generality of this result. Since the claim is usually made that quantum strategies are more powerful than classical strategies, we find that this result does not hold in our case of entangled quantum penny flip game. So the moral of the story is that quantum strategies are not(always) better than classical strategies.

The second takeaway is of relevance to the wider world of quantum computation wherein quantum algorithms have been shown to be more efficient than classical algorithms, for example Shor's algorithm. We in this work put forth a counter example which demonstrates that a particular classical algorithm can outwit the previously unbeatable quantum algorithm in the entangled quantum penny flip problem. On top of that the mixed strategy works against any possible unitary as we show by simulation on a strategy space for all possible parameters.

Finally, we show that our game is an example of "gaming the quantum" rather than "quantized games" . Our future endeavors include extending such a mixed strategy to a more general class of games, checking for the effect of noise and phenomena like decoherence in entanglement based games.

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Chapter 4

Legett Garg inequality

4.1 Introduction

- In contrast to the spatial Bell's inequalities which probe entanglement between spatially-separated systems, the Leggett-Garg inequalities test the correlations of a single system measured at different times.
- Violation of a genuine Leggett-Garg test implies either the absence of a realistic description of the system or the impossibility of measuring the system without disturbing it.
- Quantum mechanics violates the inequalities on both accounts and the original motivation for these inequalities was as a test for quantum coherence in macroscopic systems.

In the last few years quite a number of experimental tests and violations of these inequalities in a variety of microscopic systems such as superconducting qubits, nuclear spins, and photons have been shown.

Extrapolating the laws of quantum mechanics up to the scale of everyday objects, one inevitably arrives at the prospect of *macroscopic coherence*, with objects composed of very many atoms existing in quantum superpositions of macroscopically very different states. Schrödinger's cat , simultaneously both dead and alive, is the embodiment of macroscopic coherence. Of course the macroscopic world around us doesn't seem to comply with this idea !

In their 1985 paper Leggett and Garg were interested in whether macroscopic coherence could be realized in the laboratory and, if so, how one might go about demonstrating its presence. They approached this by first codifying our intuition about the macroscopic world into two principles:

(A1) *macroscopic realism* (MR) and

(A2) *Noninvasive measurability* (NIM).

MR implies that the performance of a measurement on a macroscopic system reveals a well-defined pre-existing value ("Is the flux there when nobody looks?" is thus answered in the affirmative); NIM states that, in principle, we can measure this value without disturbing the system. Whilst classical mechanics conforms with both of these assumptions, quantum mechanics certainly does not — the existence of a macroscopic superposition would violate the first, and its quantum-mechanical collapse under measurement, the second.

Based on these assumptions, Leggett and Garg went on to derive a class of inequalities that any system behaving in accord with our macroscopic intuition should obey. These are the Leggett-Garg inequalities (LGI) and they are the subject of this review. Should it be shown that a series of measurements on a system violates a LGI, then one of the above assumptions must be invalid and an intuitive macroscopic understanding of the system must be abandoned. In this way, the LGIs provide a method to investigate the existence of macroscopic coherence and to test the applicability of quantum mechanics as we scale from the micro- to the macro-scopic world.

The simplest LGI is constructed as follows. We assume that it is possible to define for the system a macroscopic dichotomic variable $Q = \pm 1$ and measure its two-time correlation functions $C_{ij} = Q(t_i)Q(t_j)$. We then perform three sets of experimental runs to measure three different C_{ij} with different pairs of time arguments. Postulates **(A1)** and **(A2)** together imply that there exists a single joint probability distribution that is

sufficient to describe all three experimental runs. From this it follows that

$$K_3 \equiv C_{21} + C_{32} - C_{31} \leq 1. \tag{4.1}$$

By considering a quantum model of a two-level system undergoing coherent oscillations between the states with $Q = \pm 1$, it is easy to show that quantum mechanics violates this inequality with a maximum value of $K_3^{\max} = 3/2$ for the two-level system.

LGIs share the same structure with, and are intimately related to, Bell's inequalities. But, whereas Bell's inequalities place bounds on correlations between measurements on spatially-separated systems, in the LGIs, the separation between measurements is in time. LGIs are for this reason often referred to as temporal Bell's inequalities . Both sets of inequalities are founded on realism, but to obtain testable inequalities that are violable by quantum mechanics, realism is cojoined with locality in the Bell's inequalities, and with NIM in the LGI. Formally, the assumptions of NIM and locality play similar roles in the derivation of the respective inequalities.

4.2 LGI violations of a qubit

Most experimental tests of LGIs to-date have been performed on two-level systems or *qubits*[2], the most elementary of quantum systems. It is thus of interest to look in-depth at the violation of the LGIs for this system. Although we consider only a very specific two-level example, it has been shown [3] that every non-trivial quantum evolution, irrespective of the nature or size of the system, allows one to violate a LGI, given the ability to make projective measurements on the initial state.

4.2.1 Maximum violations

The classical correlation functions $C_{ij} = \langle Q_i Q_j \rangle$ have no unique quantum analogue, due to issues of operator ordering. In discussing the measurement of K_n for a quantum system,

the meaning of the correlation function C_{ij} must be specified. Implicit in the original work of Leggett-Garg was that these quantities be obtained with projective measurements, in which case the correlation functions may expressed in the same way as in Eqn. 4.1. The correlators so obtained are equal to the symmetrised combination:

$$C_{ij} = \frac{1}{2}\left\langle\left\{\hat{Q}_i, \hat{Q}_j\right\}\right\rangle. \tag{4.2}$$

Parameterising the qubit operators as $\hat{Q}_i = \mathbf{a}_i \cdot \hat{\boldsymbol{\sigma}}$, with $\hat{\boldsymbol{\sigma}}$ the vector of Pauli matrices and $\mathbf{a}_i$ a unit vector, and using the identity $(\mathbf{a}_2 \cdot \hat{\boldsymbol{\sigma}})(\mathbf{a}_3 \cdot \hat{\boldsymbol{\sigma}}) = \mathbf{a}_2 \cdot \mathbf{a}_3\ \hat{\mathbb{1}} + i\hat{\boldsymbol{\sigma}} \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$, we obtain

$$\tfrac{1}{2}\left\langle\left\{\hat{Q}_i, \hat{Q}_j\right\}\right\rangle = \mathbf{a}_i \cdot \mathbf{a}_j \left\langle\hat{\mathbb{1}}\right\rangle = \mathbf{a}_i \cdot \mathbf{a}_j, \tag{4.3}$$

independent of initial conditions. The nth-order Leggett-Garg parameter then reads $K_n = \sum_{m=1}^{n-1} \mathbf{a}_{m+1} \cdot \mathbf{a}_m - \mathbf{a}_n \cdot \mathbf{a}_1$.

Violations of the LGIs can be associated with the non-commutativity of the operator $\hat{Q}$ with itself at different times. With the above parameterisation, we have the commutation relation

$$\left[\hat{Q}_i, \hat{Q}_j\right] = 2i\hat{\boldsymbol{\sigma}} \cdot (\mathbf{a}_i \times \mathbf{a}_j). \tag{4.4}$$

Assuming that the vectors $\mathbf{a}_i$ all lie in the x-z plane with equal angles between them, $\theta_i = \theta$, the commutators between relevant operator pairs are

$$\left[\hat{Q}_2, \hat{Q}_1\right] = \left[\hat{Q}_3, \hat{Q}_2\right] = 2i\hat{\sigma}_y \sin\theta; \quad \text{and} \qquad \left[\hat{Q}_3, \hat{Q}_1\right] = 2i\sigma_y \sin 2\theta. \tag{4.5}$$

The points where these commutators simultaneously vanish are the points where violations of the LGIs (K_3 and K_3' together) disappear. Furthermore, the sum of the magnitudes of these commutators is proportional to $2|\sin\theta| + |\sin 2\theta|$, which is maximized by setting $\theta_1 = \theta_2 = \pm\pi/3$. Thus the points where the commutators are simultaneously maximized are the points where the LGI violations are greatest.

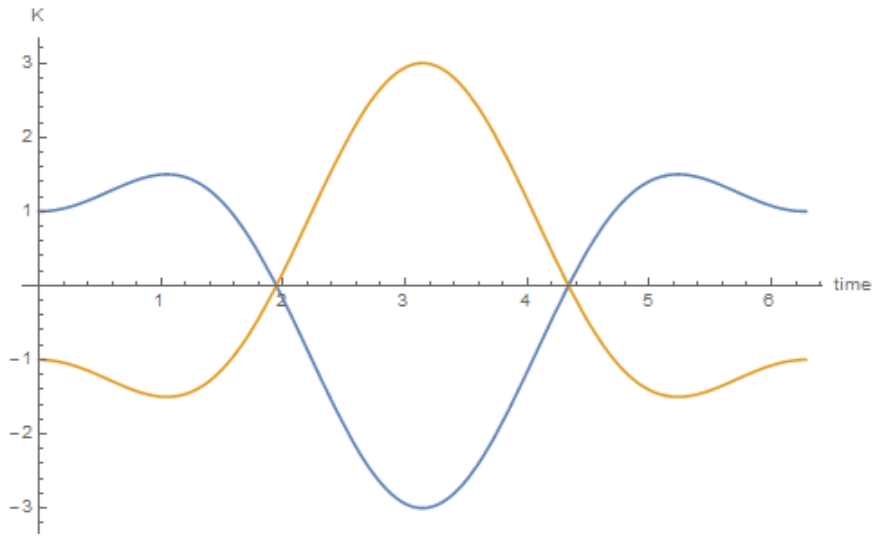

Figure 4.1: Third-order Leggett-Garg function K_3 with equi-spaced measurements for a qubit as a function of measurement-time spacing τ. A violation of one or the other of the K_3 and K_3' inequalities occurs for all τ except at multiples of $\pi/2$.

4.2.2 Time evolution

The canonical example of a time evolution that violates the LGIs is a qubit evolving under the Hamiltonian

$$\hat{H}_{\mathrm{qb}} = \frac{1}{2}\Omega\hat{\sigma}_x, \tag{4.6}$$

and measured in the z-direction, $\hat{Q} = \hat{\sigma}_z$. In this case, the correlation functions read [1]

$$C_{ij} = \cos\Omega(t_i - t_j), \tag{4.7}$$

and choosing equal time intervals, $t_{m+1} - t_m = \tau$, we obtain:

$$K_n = (n-1)\cos\Omega\tau - \cos(n-1)\Omega\tau. \tag{4.8}$$

The third-order K_3 is plotted in Fig. 4.1. It oscillates as a function of the measurement time τ with maximum value of $3/2$ occurring at times $\Omega\tau = \pm\frac{\pi}{3} + 2\pi k$, with k an integer. Only for certain ranges of τ is $K_3 > 1$. At third order, permutation of the time indices

only either recovers the original inequality, Eq. (4.8), or generates the trivially-satisfied $\cos 2\Omega\tau < 1$. The K_3' inequality of however, yields the distinct

$$-3 \leq -2\cos(\Omega\tau) - \cos(2\Omega\tau) \leq 1. \tag{4.9}$$

K_3' has maxima of $3/2$ at $\Omega\tau = \pm\frac{2\pi}{3} + 2\pi k$ and, as Fig. 4.1 shows, is complementary to K_3 in that the violations of K_3' fill in the gaps between those of K_3. The only times for which no violation occurs is when $\Omega\tau = \frac{k}{2}\pi$, where the system state is an eigenstate of the measurement operator and a QND measurement is performed.

4.3 Future plans

We wish to use Legett-Garg inequalities to explore two aspects:

- To explore the possibility of quantum state tomography using the correlations
- Using LG inequality to show a quantum to classical transition in games

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Chapter 5

Quantum Walks

5.1 Introduction

What is a random walk?

A random walk is a mathematical formalization of a path that consists of a succession of random steps.

- For example, the path traced by a molecule as it travels in a liquid or a gas, the search path of a foraging animal, the price of a fluctuating stock and the financial status of a gambler can all be modeled as random walks, although they may not be truly random in reality.

- Random walks have been used in many fields: ecology, economics, psychology, computer science, physics, chemistry, and biology. Random walks explain the observed behaviors of many processes in these fields, and thus serve as a fundamental model for the recorded stochastic activity.

- Often, random walks are assumed to be Markov chains. A Markov chain (discrete-time Markov chain or DTMC), is a random process that undergoes transitions from one state to another on a state space. It must possess a property that is usually characterized as ''memorylessness'': the probability distribution of the next state depends only on the current state and not on the sequence of events that preceded it.

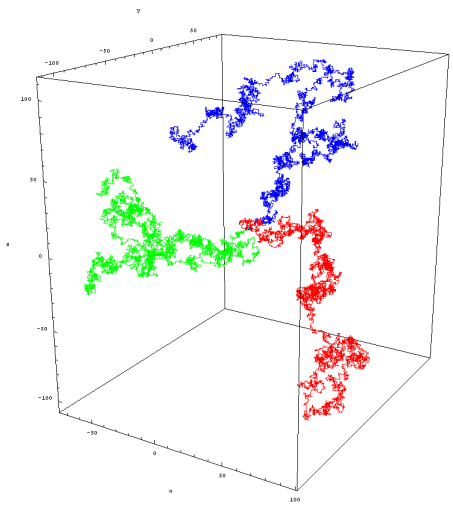

Figure 5.1: 3 random walkers in 3D. Source: Wikipedia

5.2 Quantum random walks: A gentle introduction

- Imagine a particle on a line whose position is described by a wave-packet $|\psi_{x_0}\rangle$ localized around a position x_0, i.e. the function $\langle x|\psi_{x_0}\rangle$ corresponds to a wave-packet centered around x_0. Let P be the momentum operator. The translation of the particle, corresponding to a step of length l can be represented by the unitary operator $U_l = exp(-iPl/\hbar)$, so that $U_l|\psi_{x_0}\rangle = |\psi_{x_0-l}\rangle$. Now let us also assume that the particle has a spin-$\frac{1}{2}$ degree of freedom. Let S_z represent the operator corresponding to the z component of the spin and denote the eigenstates of S_z by $|\uparrow\rangle$ and $|\downarrow\rangle$, so that $S_z|\uparrow\rangle = \frac{\hbar}{2}|\uparrow\rangle$ and $S_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle$. From now on we will set $\hbar = 1$ to simplify the notation.

A spin-$\frac{1}{2}$ particle is usually described by a 2-vector $|\Psi\rangle = (|\tilde{\psi}^{\uparrow}\rangle, |\tilde{\psi}^{\downarrow}\rangle)^T$, where the first part is the component of the wave-function of the particle in the spin- $|\uparrow\rangle$ space and the second one is the component in the $|\downarrow\rangle$-space. Normalization requires that $\||\tilde{\psi}^{\uparrow}\rangle\|^2 + \||\tilde{\psi}^{\downarrow}\rangle\|^2 = 1$. To emphasize the tensor-structure of the space of the particle we will write this in a slightly different but equivalent way as $|\Psi\rangle = \alpha^{\uparrow}|\uparrow\rangle \otimes |\psi^{\uparrow}\rangle + \alpha^{\downarrow}|\downarrow$

$\rangle \otimes |\psi^{\downarrow}\rangle$, where we normalize the two wave-functions $\langle\psi^{\uparrow}|\psi^{\uparrow}\rangle = \langle\psi^{\downarrow}|\psi^{\downarrow}\rangle = 1$, so that $|\alpha^{\uparrow}|^2 + |\alpha^{\downarrow}|^2 = 1$. The tensor product '$\otimes$' separates the two degrees of freedom, spin and space, and will allow us to view the resulting correlations between these two degrees of freedom more clearly. The time development corresponding to a translation by l on the larger state-space of the spin-$\frac{1}{2}$ particle can now be described by the unitary operator $U = exp(-2iS_z \otimes Pl)$.

This operator induces a kind of conditional translation of the particle depending on its internal spin-degree of freedom. In particular if the spin of the particle is initially in the state $|\uparrow\rangle$, so that its wave-function is of the form $|\uparrow\rangle \otimes |\psi^{\uparrow}_{x_0}\rangle$, then application of U transforms it to $|\uparrow\rangle \otimes |\psi^{\uparrow}_{x_0-l}\rangle$ and the particle will be shifted to the right by l. If the spin of the particle is in the state $|\downarrow\rangle$, i.e. the total wave-function is given by $|\downarrow\rangle \otimes |\psi^{\downarrow}_{x_0}\rangle$, then the translation operator will transform it to $|\downarrow\rangle \otimes |\psi^{\downarrow}_{x_0+l}\rangle$ and the particle will be shifted to the left. More interesting behavior occurs when the initial spin state of the particle, localized in x_0, is not in an eigenstate of S_z, but rather in a superposition

$$|\Psi_{in}\rangle = (\alpha^{\uparrow}|\uparrow\rangle + \alpha^{\downarrow}|\downarrow\rangle) \otimes |\psi_{x_0}\rangle. \tag{5.1}$$

Application of the translation operator U will induce a superposition of positions

$$U|\Psi_{in}\rangle = \alpha^{\uparrow}|\uparrow\rangle \otimes |\psi_{x_0-l}\rangle + \alpha^{\downarrow}|\downarrow\rangle \otimes |\psi_{x_0+l}\rangle. \tag{5.2}$$

5.3 Connection to classical random walks

If at this point we decide to *measure* the spin in the S_z basis, the particle will be either in the state $|\uparrow\rangle \otimes |\psi_{x_0-l}\rangle$, localized around $x_0 + l$ with probability $p^{\uparrow} = |\alpha^{\uparrow}|^2$ or in the state $|\downarrow\rangle \otimes |\psi_{x_0-l}\rangle$, localized around $x_0 - l$ with probability $p^{\downarrow} = |\alpha^{\downarrow}|^2$. This procedure corresponds to a (biased) random walk of a particle on the line: we can imagine that a biased coin with probabilities of head/tail $p^{\uparrow}/p^{\downarrow}$ is flipped. Upon head the particle moves

right and upon tail the particle moves left. After this step the particle is on average displaced by $l(p^{\uparrow} - p^{\downarrow})$. If we repeat this procedure T times (each time measuring the spin in the basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ and re-initializing the spin in the state $\alpha^{\uparrow}|\uparrow\rangle + \alpha^{\downarrow}|\downarrow\rangle$), the particle will be displaced on average by an amount $\langle x\rangle = Tl(p^{\uparrow} - p^{\downarrow}) = Tl(|\alpha^{\uparrow}|^2 - |\alpha^{\downarrow}|^2)$ and the variance of its distribution on the line will be $\sigma^2 = 4Tl|\alpha^{\uparrow}|^2|\alpha^{\downarrow}|^2 = 4Tl^2p^{\uparrow}p^{\downarrow}$. This is exactly what we obtain if the particle performs a (biased) random walk on the line. We can imagine that the spin of the particle takes the role of a coin. This coin is 'flipped' and the outcome determines the direction of the step in the random walk of the particle.

Now let us consider the following modification of this procedure. Instead of measuring the spin in the eigenbasis of S_z we will measure it in some rotated basis, given by two orthogonal vectors $\{|s_+\rangle, |s_-\rangle\}$. Alternatively we can *rotate* the spin by some angle θ before measuring it in the S_z eigenbasis. Let us be more formal and set up some more of the language used in quantum information theory. If we identify

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{5.3}$$

we can write

$$S_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|). \tag{5.4}$$

A rotation of the spin can be described by the matrix[1]

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{5.5}$$

Let us denote the measurement in the S_z basis by M_z. To explore the effect of the suite of operations $M_z R(\theta) U$ on the initial state $|\Psi_{in}\rangle$ we have to slightly rewrite the operator U:

$$\begin{aligned} U &= e^{-2iS_z \otimes Pl} = e^{-i(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|)\otimes Pl} \\ &= (|\uparrow\rangle\langle\uparrow| \otimes e^{-iPl})(|\downarrow\rangle\langle\downarrow| \otimes e^{iPl}). \end{aligned} \tag{5.6}$$

[1]Note that this is not the most general unitary transformation on a two-level system. A general (normalized) unitary is given by two parameters (θ, ϕ).

The second equality follows from the fact that $|\uparrow\rangle\langle\uparrow| \otimes Pl$ and $|\downarrow\rangle\langle\downarrow| \otimes Pl$ commute[2] together with the observation that $|\uparrow\rangle\langle\uparrow|$ and $|\downarrow\rangle\langle\downarrow|$ are projectors, so $|\uparrow\rangle\langle\uparrow|^k = |\uparrow\rangle\langle\uparrow|$ and $|\downarrow\rangle\langle\downarrow|^k = |\downarrow\rangle\langle\downarrow|$ when we expand the exponential. Now

$$U|\Psi_{in}\rangle = (\alpha^{\uparrow}|\uparrow\rangle \otimes e^{-iPl} + \alpha^{\downarrow}|\downarrow\rangle \otimes e^{iPl})|\psi_{x_0}\rangle \tag{5.7}$$

and applying $R(\theta)$ from the left gives

$$\begin{aligned}&\big[(\alpha^{\uparrow}\cos\theta e^{-iPl} - \alpha^{\downarrow}\sin\theta e^{iPl})|\uparrow\rangle \\ &+(\alpha^{\uparrow}\sin\theta e^{-iPl} + \alpha^{\downarrow}\cos\theta e^{iPl})|\downarrow\rangle\big] \otimes |\psi_{x_0}\rangle.\end{aligned} \tag{5.8}$$

If the width Δx of the initial wave packet is much larger than the step length l we can approximate $\exp(\pm iPl)|\psi_{x_0}\rangle \approx (I \pm iPl)|\psi_{x_0}\rangle$. With this in mind we can establish the state of the particle after the measurement M_z:

$$M_z R(\theta) U|\Psi_{in}\rangle = \begin{cases} |\uparrow\rangle \otimes (I - iPl\delta^{\uparrow})|\psi_{x_0}\rangle \\ |\downarrow\rangle \otimes (I - iPl\delta^{\downarrow})|\psi_{x_0}\rangle \end{cases} \tag{5.9}$$

with probabilities

$$p^{\uparrow} = |\alpha^{\uparrow}\cos\theta - \alpha^{\downarrow}\sin\theta|^2 \quad p^{\downarrow} = |\alpha^{\uparrow}\sin\theta + \alpha^{\downarrow}\cos\theta|^2 \tag{5.10}$$

and displacements

$$l\delta^{\uparrow} := l\frac{\alpha^{\uparrow}\cos\theta + \alpha^{\downarrow}\sin\theta}{\alpha^{\uparrow}\cos\theta - \alpha^{\downarrow}\sin\theta} \quad l\delta^{\downarrow} := l\frac{\alpha^{\uparrow}\sin\theta - \alpha^{\downarrow}\cos\theta}{\alpha^{\uparrow}\sin\theta + \alpha^{\downarrow}\cos\theta}. \tag{5.11}$$

If the width Δx of the initial wave packet is much larger than $l\delta^{\uparrow}$ and $l\delta^{\downarrow}$, we can again approximate $(I - iPl\delta^{\uparrow,\downarrow})|\psi_{x_0}\rangle \approx \exp(-iPl\delta^{\uparrow,\downarrow})|\psi_{x_0}\rangle = |\psi_{x_0 - l\delta^{\uparrow,\downarrow}}\rangle$.

Now it is crucial to note that the displacement of the particle in one of the two cases, $l\delta^{\uparrow}$ say, can be made much *larger* than l. For instance we may choose $\tan\theta = |\alpha^{\uparrow}/\alpha^{\downarrow}|(1+\epsilon)$ with $l/\Delta x \ll |\epsilon| \ll 1$. Then the displacement of the particle in case we measured $|\uparrow\rangle$ will be $l\delta^{\uparrow} \approx -2l/\epsilon$ which can be several orders of magnitude larger than

[2] The equality $\exp(A+B) = \exp(A)\exp(B)$ is true if A and B commute, i.e. $AB = BA$.

l. However this event is 'rare': the probability to measure $|\uparrow\rangle$ is $p^{\uparrow} \approx |\alpha^{\uparrow}\alpha^{\downarrow}|^2\epsilon^2$. For the same choice of parameters the probability to measure $|\downarrow\rangle$ is $p^{\downarrow} \approx 1 - |\alpha^{\uparrow}\alpha^{\downarrow}|^2\epsilon^2$ and the displacement in that event is $l\delta^{\downarrow} \approx l(|\alpha^{\uparrow}|^2 - |\alpha^{\downarrow}|^2) + O(l\epsilon)$. For the average displacement $p^{\uparrow}l\delta^{\uparrow} + p^{\downarrow}l\delta^{\downarrow}$ we get as before $l(|\alpha^{\uparrow}|^2 - |\alpha^{\downarrow}|^2)$, and similarly the variance of the walk will be unchanged. We have the effect that even though our translation operation displaces only by l, in some (rare) cases the particle will jump much further than l. The effect is illustrated in the Figure below.

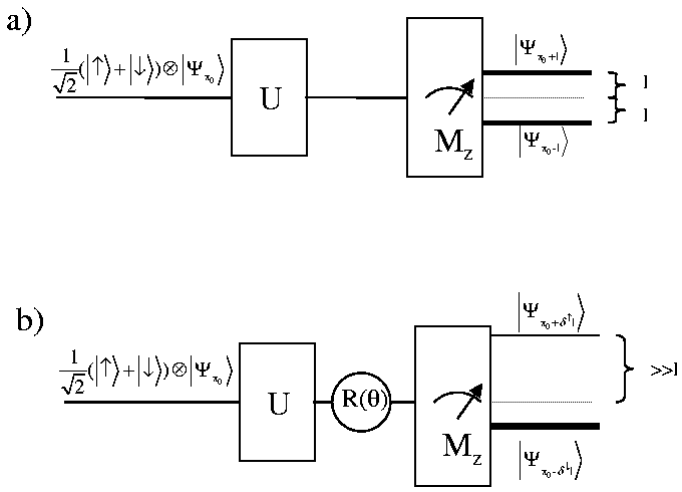

Figure 5.2: Classical vs Quantum random walk. *This figure is reproduced from Ref [1] with acknowledgement for explanatory purposes*

5.4 Key ideas

The key idea behind quantum random walks is to iterate the walk we have introduced in the previous section *without* resorting to intermediate measurements. In other words we will repeat the succession of unitary translation U and rotation $R(\theta)$ without the measurement M_z at intermediate time steps. Furthermore we will discretize the position space of the particle, or in other words we put the particle on a lattice or a graph. Usually this space will be finite. A discrete and finite state space is important in the context of

simulability on a (finite) computer. A quantum computer works with discrete registers, i.e. its state-space is a Hilbert space of large but finite dimension. Discretizing the quantum walk will allow us to map it to a computation by such a machine.

5.5 The discrete model

Let $\mathcal{H}_P$ be the Hilbert space spanned by the positions of the particle. For a line with grid-length 1 this space is spanned by basis states $\{|i\rangle : i \in \mathbf{Z}\}$; if we work on a circle of size N we have $\mathcal{H}_P = \{|i\rangle : i = 0 \dots N-1\}$. States in this basis will take the role of the wave function $|\psi\rangle$ of previous section with $|i\rangle$ corresponding to a particle localized in position i. We will not be concerned any more about the width of the distribution of the wave function - our model describes unitary transformations of states of a finite Hilbert space $\mathcal{H}$ and the notion of a particle is only used to guide our intuition from now on. Yet later, when we talk about the physical implementation of the quantum random walk, we will see that we can approximate the states of this mathematical model by particles (atoms, photons, etc.) with wave functions of finite width.

The position Hilbert space $\mathcal{H}_P$ is augmented by a 'coin'-space $\mathcal{H}_C$ spanned by two basis states $\{|\uparrow\rangle, |\downarrow\rangle\}$, which take the role of the spin-$\frac{1}{2}$ space in the previous section. States of the total system are in the space $\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_P$ as before. The conditional translation of the system (taking the role of $\exp(-2iS_z \otimes Pl)$) can be described by the following unitary operation

$$S = |\uparrow\rangle\langle\uparrow| \otimes \sum_i |i+1\rangle\langle i| + |\downarrow\rangle\langle\downarrow| \otimes \sum_i |i-1\rangle\langle i| \tag{5.12}$$

where the index i runs over Z in the case of a line or $0 \leq i \leq N-1$ in the case of a circle. In the latter case we always identify 0 and N, so all arithmetic is performed modulo N. S transforms the basis state $|\uparrow\rangle \otimes |i\rangle$ to $|\uparrow\rangle \otimes |i+1\rangle$ and $|\downarrow\rangle \otimes |i\rangle$ to $|\downarrow\rangle \otimes |i-1\rangle$.

The first step of the random walk is a rotation in the coin-space, which in analogy to

the classical random walk we will call 'coin-flip' C. The unitary transformation C is very arbitrary and we can define a rich family of walks with different behavior by modifying C. For the moment we will want C to be 'unbiased' in the following sense. Assume we would initialize the walk in a localized state $|0\rangle$ with the coin in one of the basis states, say $|\uparrow\rangle$. If we chose to measure the coin-register of the walk in the (standard) basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ after one iteration (coin-flip C followed by translation S) we want to obtain the classical probability distribution of the unbiased walk, namely a translation to the right ($|1\rangle$) with probability $\frac{1}{2}$ and a step to the left ($|-1\rangle$) otherwise. A frequently used balanced unitary coin is the so called Hadamard coin H

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{5.13}$$

It is easy to see that the Hadamard coin is balanced:

$$\begin{aligned} |\uparrow\rangle \otimes |0\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\ &\xrightarrow{S} \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |1\rangle + |\downarrow\rangle \otimes |-1\rangle). \end{aligned} \tag{5.14}$$

Measuring the coin state in the standard basis gives each of $\{|\uparrow\rangle \otimes |1\rangle, |\downarrow\rangle \otimes |-1\rangle\}$ with probability $1/2$. After this measurement there is no correlation between the positions left. If we continued the quantum walk with such a measurement at each iteration we obtain the plain classical random walk on the line (or on the circle). The distribution of this walk can be well illustrated with the help of Galton's board (also called Quincunx) . Its limiting distribution on the line (for large number of iterations T) approaches a Gaussian distribution with mean zero and variance $\sigma^2 = T$. On the circle this distribution approaches the uniform distribution.

The quantum random walk of T steps is defined as the transformation U^T, where U, acting on $\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_P$ is given by

$$U = S \cdot (C \otimes I). \tag{5.15}$$

$T \backslash i$	-5	-4	-3	-2	-1	0	1	2	3	4	5
0						1					
1					$\frac{1}{2}$		$\frac{1}{2}$				
2				$\frac{1}{4}$		$\frac{1}{2}$		$\frac{1}{4}$			
3			$\frac{1}{8}$		$\frac{3}{8}$		$\frac{3}{8}$		$\frac{1}{8}$		
4		$\frac{1}{16}$		$\frac{1}{4}$		$\frac{3}{8}$		$\frac{1}{4}$		$\frac{1}{16}$	
5	$\frac{1}{32}$		$\frac{5}{32}$		$\frac{5}{16}$		$\frac{5}{16}$		$\frac{5}{32}$		$\frac{1}{32}$

Figure 5.3: The probability of being at position i after T steps of the classical random walk on the line starting in 0.

$T \backslash i$	-5	-4	-3	-2	-1	0	1	2	3	4	5
0						1					
1					$\frac{1}{2}$		$\frac{1}{2}$				
2				$\frac{1}{4}$		$\frac{1}{2}$	0	$\frac{1}{4}$			
3			$\frac{1}{8}$		$\frac{5}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		
4		$\frac{1}{16}$		$\frac{5}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{16}$	
5	$\frac{1}{32}$		$\frac{17}{32}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{5}{32}$		$\frac{1}{32}$

Figure 5.4: The probability of being found at position i after T steps of the quantum random walk on the line, with the initial state $|\Phi_{in}\rangle = |\downarrow\rangle \otimes |0\rangle$. Note that this distribution starts to differ from the classical distribution from $T = 3$ on. Furthermore the quantum random walk is asymmetric with a drift to the left.

To illustrate the departure of the quantum walk from its classical counterpart let us evolve the walk (without intermediate measurements) for some steps starting in the initial state $|\Phi_{in}\rangle = |\downarrow\rangle \otimes |0\rangle$ and study the induced probability distribution on the positions.

$$\begin{aligned}
|\Phi_{in}\rangle &\xrightarrow{U} \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |1\rangle - |\downarrow\rangle \otimes |-1\rangle) \\
&\xrightarrow{U} \frac{1}{2}(|\uparrow\rangle \otimes |2\rangle - (|\uparrow\rangle - |\downarrow\rangle) \otimes |0\rangle + |\downarrow\rangle \otimes |-2\rangle) \\
&\xrightarrow{U} \frac{1}{2\sqrt{2}}(|\uparrow\rangle \otimes |3\rangle + |\downarrow\rangle \otimes |1\rangle + |\uparrow\rangle \otimes |-1\rangle \\
&\qquad -2|\downarrow\rangle \otimes |-1\rangle - |\downarrow\rangle \otimes |-3\rangle).
\end{aligned} \tag{5.16}$$

This example already shows that the probability distribution induced by the quantum walk differs from the classical one.

5.6 Asymmetry

A first thing to notice is that the quantum random walk induces an asymmetric probability distribution on the positions, it is 'drifting' to the left. This asymmetry arises from the fact that the Hadamard coin treats the two directions $|\uparrow\rangle$ and $|\downarrow\rangle$ differently; it multiplies the phase by -1 only in the case of $|\downarrow\rangle$. Intuitively this induces more cancellations for paths going right-wards (destructive interference), whereas particles moving to the left interfere constructively.

There are two ways to mend this asymmetry. Inspecting Eq. (5.14) which describes the first step of the Hadamard walk starting in $|\uparrow\rangle \otimes |0\rangle$ (instead of in $|\Phi_{in}\rangle = |\downarrow\rangle \otimes |0\rangle$ as in Eq. (5.16)) and iterating the walk in Eq. (5.14) more we see that the walk starting in $|\uparrow\rangle \otimes |0\rangle$ has a drift to the right side, exactly opposite to the walk in Eq. (5.16). To obtain a symmetric distribution we can start the walk in a superposition of $|\uparrow\rangle$ and $|\downarrow\rangle$ and make sure that these two drifts do not interfere with each other. This can be achieved starting in the symmetric state $|\Phi_{sym}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle) \otimes |0\rangle$. Since the Hadamard walk does not introduce any complex amplitudes the trajectories from $|\uparrow\rangle$ will stay real and the ones from $|\downarrow\rangle$ will be purely imaginary; they will hence not interfere with each other, making the total distribution symmetric.

Another solution to eliminate asymmetry of the walk is to use a different (balanced) coin, namely

$$Y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}. \tag{5.17}$$

It is not hard to see that this coin treats $|\uparrow\rangle$ and $|\downarrow\rangle$ in the same way and does not bias the walk, independently of its initial coin-state.

As mentioned before the classical symmetric random walk on the line after T steps has a variance $\sigma^2 = T$, so the expected distance from the origin is of order $\sigma = \sqrt{T}$. By contrast it can be shown that the quantum random walk has a variance that scales with

$\sigma^2 \sim T^2$, which implies that the expected distance from the origin is of order $\sigma \sim T$ - the quantum walk propagates quadratically faster!

5.7 Random walks on general Graphs

d-regular graphs are graphs with vertex-degree d, i.e. each vertex has d outgoing edges. The coin Hilbert space $\mathcal{H}_C$ is of dimension d. For every vertex let us internally label all the outgoing edges with $j \in 1 \dots d$ such that each edge has a distinct label. Let us call e_v^j an edge $e = (v, w)$ which on v's end is labeled by j. The state associated with a vertex v pointing along an edge labeled j is $|j\rangle \otimes |v\rangle$ (corresponding to $|\uparrow\rangle \otimes |i\rangle$ on the line or circle). Now we can define a conditional shift operation S

$$S|j\rangle \otimes |v\rangle = \begin{cases} |j\rangle \otimes |w\rangle & \text{if } e_v^j = (v, w) \\ 0 & \text{o/w} \end{cases} \tag{5.18}$$

S moves the particle from v to w if the edge (v, w) is labeled by j on v's side. If the graph is not d-regular we can still define S in the same way if we let d be the maximum degree of any vertex in the graph. Figure below shows a graph and a valid labeling.

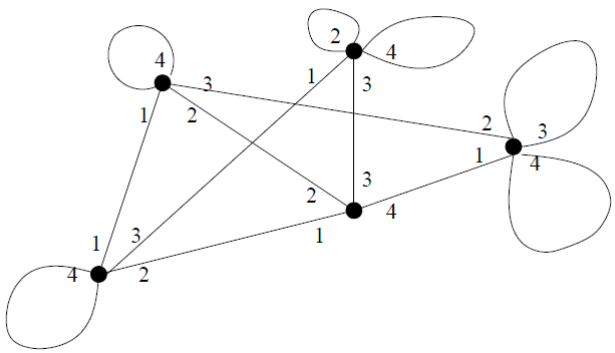

Figure 5.5: A graph with various degrees and a labeling of the edges for each vertex. Note that the label associated with and edge (v, w) can be different from v's end and from w's end. *This figure is reproduced from Ref [1] with acknowledgement for explanatory purposes*

If we wish to retain the property in general graphs that the coin is balanced (i.e. every direction is obtained with equal probability if we measure the coin space) we can use the following coin which generalizes H

$$DFT = \frac{1}{\sqrt{d}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{d-1} \\ & & \dots & & \\ 1 & \omega^{d-1} & \omega^{2(d-1)} & \dots & \omega^{(d-1)(d-1)} \end{pmatrix}. \tag{5.19}$$

Here $\omega = \exp(2\pi i/d)$ is a d-th root of unity. Clearly the unitary DFT-coin[3] transforms each direction into an equally weighted superposition of directions such that after measurement each of them is equally likely to be obtained (with probability $1/d$).

5.8 An example: S-T Connectivity

Let us illustrate how random walks are harnessed in classical computer science by describing how they help to solve the $s-t$ connectivity problem. Given an undirected graph G with vertex set V of size $|V|$, the problem is to decide whether two vertices s and t are connected or not. This problem is a natural abstraction of a number of graph search problems and is important in the study of space bounded complexity classes in computer science.

It is easy to see that a standard graph search algorithm that keeps track of all the vertices the search has visited can do the job in a time linear in the number of edges of the graph. Let us assume now that our computer is limited in memory and can keep track only of a few vertices at a time[4]. It can perform a random walk on the vertices of the graph starting in s. At each step of the walk the computer choses one of the adjacent

[3]The name of the coin 'DFT' comes from 'Discrete Fourier Transform'. This transformation is a cornerstone in quantum algorithms and a building block for algorithms like factoring. It has the property that it is efficiently implementable on a quantum computer.

[4]More formally we can define a probabilistic Turing machine with logarithmic space. The class of languages this Turing machine can decide in polynomial time is called RLP. Random walk arguments show that $s-t$ connectivity is in RLP

edges with equal probability and moves to the vertex that is connected to the current position by that edge. If the computer reaches t in the course of the walk it outputs YES. If it never sees t after $T = |V|^3$ steps it outputs NO.

Clearly the machine will never output YES when s and t are not in the same connected component. It turns out that the simple random walk on a graph with $|V|$ vertices has at least probability $1/2$ to hit any given vertex starting from any other in $|V|^3$ steps. This means that our machine will err with probability $1/2$ when it outputs NO. We can make this probability arbitrarily small by repeating the algorithm several times. Assume that s and t are connected. Then the error probability of an algorithm that iterates the above procedure k times is simply the probability to output NO k times. Since we assume that each iteration is independent of the previous one, this probability is given by $(1/2)^k$. If we want to make this probability very small (ϵ) we need to repeat the algorithm roughly $k = \log(\frac{1}{\epsilon})$ times.

5.9 Physical Implementation

The physical system is an optical cavity traversed by an atom. During the traversal the internal electronic levels of the atom interact with the cavity mode. Furthermore the internal atomic levels can be addressed and manipulated with a periodic sequence of microwave pulses. The focus of this proposal is to implement the random walk with existing present day technology in a microwave cavity. In this set-up the position of the random walk is encoded into a single cavity mode, whereas the coin states are the internal atomic states, when pointing out a first 'quantum random walk effect' in the original 1993 work , already suggested to observe it in the optical cavity setting. The atom in an excited state traverses the cavity. The excited internal atomic state couples to the cavity mode. If upon exit the atom is measured in a rotated basis and a rare event is observed, the wave

function of the cavity mode (initially in a coherent state with mean photon number N, for instance) can suddenly shift by much more than (classically) expected. As described the mean photon number can shift to $N + \delta$ (instead of $N + 1$) with $1 \ll \delta$. Here a random walk on the circle is achieved.

5.10 Decoherence

The crucial difference between the quantum and the classical walk are the quantum coherences, which are present only in the first. So a transition from quantum to classical has to somehow kill these coherences. We have seen that the classical walk can be obtained from the quantum walk if we measure the coin register at every iteration. Another way to obtain the classical walk is to measure the positions at every step. We can now imagine some continuous way to interpolate between non-measuring (the quantum case) and completely measuring (the classical case) and study the transition along this pathway. An intermediate step would be to measure the register (coin or position or both) only with some probability p_{meas} at each step, where $p_{meas} = 0$ corresponds to the purely quantum and $p_{meas} = 1$ to the purely classical case. Numerical studies of this transition from quantum to classical have been executed for the circle, line and hypercube by Kendon and Tregenna and extensive analytical expressions for the walk on the line with decoherence of the coin-space have been obtained by Brun et al.

5.11 Future plans

We wish to explore the merger or quantum walks and quantum games

References

[1] Kempe, Julia. "Quantum random walks: an introductory overview." Contemporary Physics 44.4 (2003): 307-327.

Chapter 6

Quantum Walks as a Quantum Game

6.1 Introduction

Here we study the role of pure and mixed quantum strategies in a 2-player entangled quantum game which manifesting itself as a quantum random walk with entangled coins. The game is played between a player A with access to quantum strategies while player B is restricted to classical pure/mixed strategies. We contrast the behavior of pure quantum strategies with that of mixed classical strategies.

6.2 Initial state and unitary dependence of quantum walks

Consider a one-dimensional quantum random walk on a line. The initial state of the coin is(say) $|0\rangle$ at position x=0 i.e.:

$$\psi = |0_c\rangle \otimes |0_x\rangle \tag{6.1}$$

Both the players share the same shift operator, S:

$$\hat{S} = |0\rangle_c\langle 0| \otimes \sum_i |i-1\rangle_p\langle i| + |1\rangle_c\langle 1| \otimes \sum_i |i+1\rangle_p\langle i| \tag{6.2}$$

which means if the spin is up or zero, the particle moves to the left and if the spin is down or one, the spin moves to the right.

Consider a simple Hadamard walk with initial state zero i.e. $|0\rangle$.

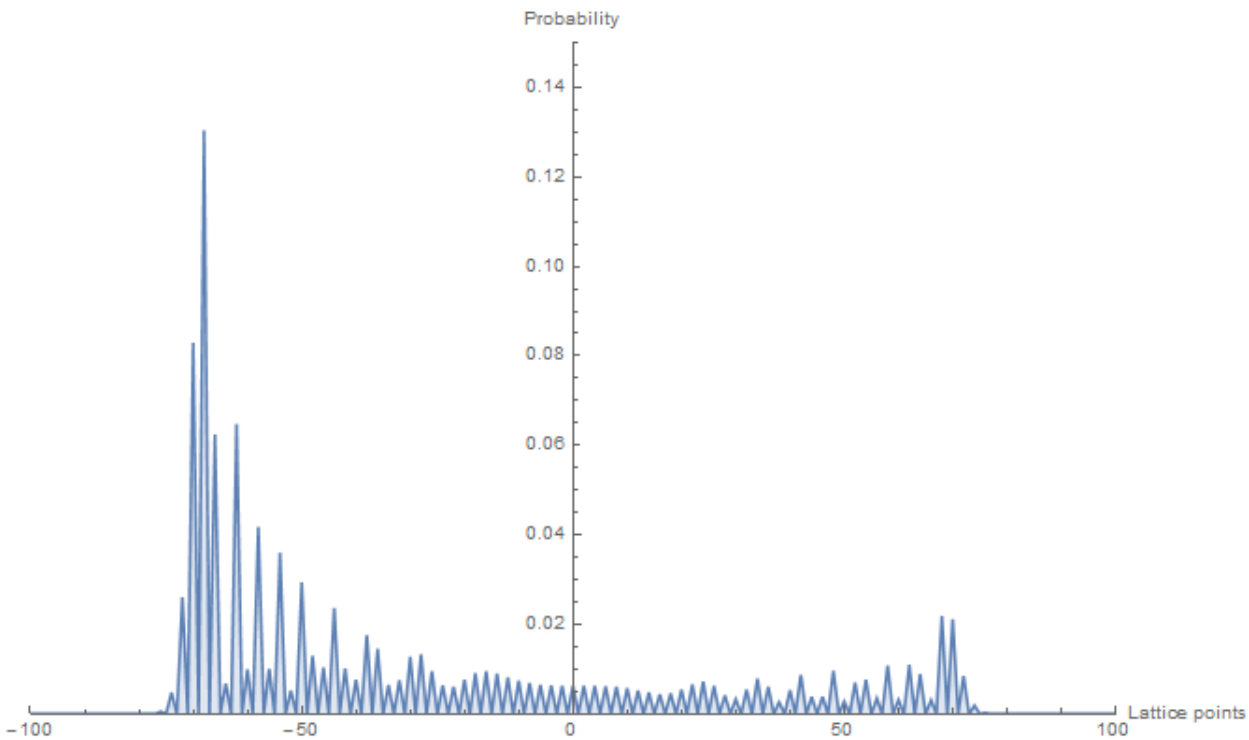

Figure 6.1: Hadamard walk with initial state $|0\rangle$ and skewness to the left.

Contrast this with a Hadamard walk with initial state one i.e. $|0\rangle$.

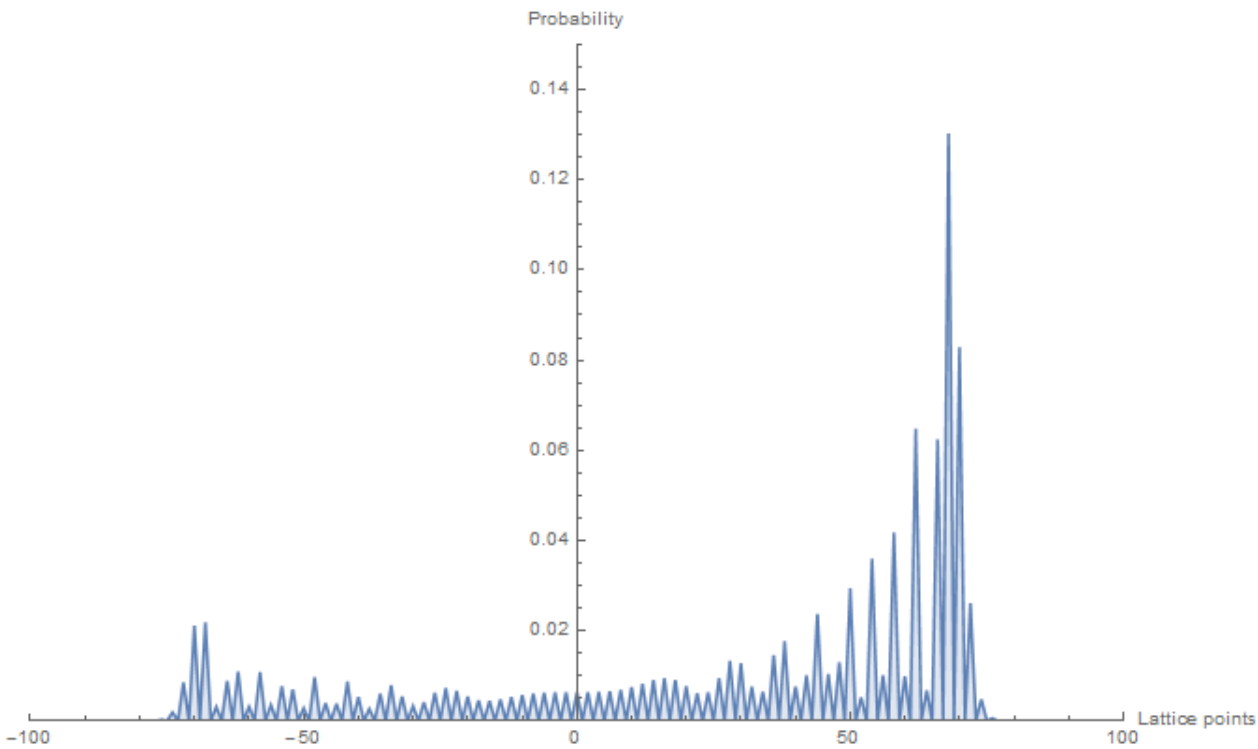

Figure 6.2: Hadamard walk with initial state $|1\rangle$ and skewness to the right.

The Hadamard is a unitary operator defined as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{6.3}$$

From the form of the operator, and the skewness of the distributions one can see that the walk depends on the initial state of the coin. This is because the coin operator namely Hadamard is not a symmetric operator w.r.t the initial state $|0\rangle$ and $|1\rangle$.
To make Hadamard symmetric, one can change the initial state to $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

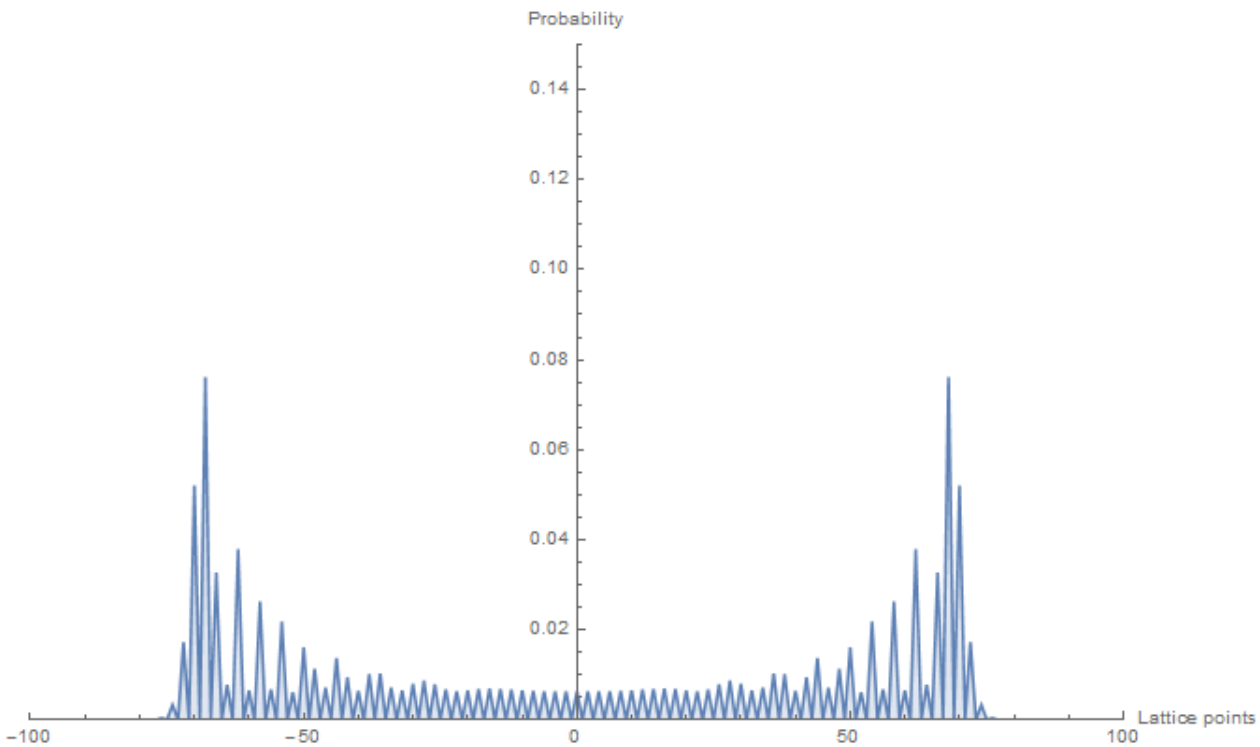

Figure 6.3: Hadamard walk with initial state $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and symmetric distribution.

which results in a symmetric distribution.

Using a more general unitary of the form

$$B = B_{\xi,\theta,\zeta} \equiv \begin{pmatrix} e^{i\xi}\cos(\theta) & e^{i\zeta}\sin(\theta) \\ e^{-i\zeta}\sin(\theta) & -e^{-i\xi}\cos(\theta) \end{pmatrix} \in U(2), \tag{6.4}$$

on the initial state $\frac{1}{2}(|0\rangle + i|1\rangle)$, we get for the tuple (30,30,0).

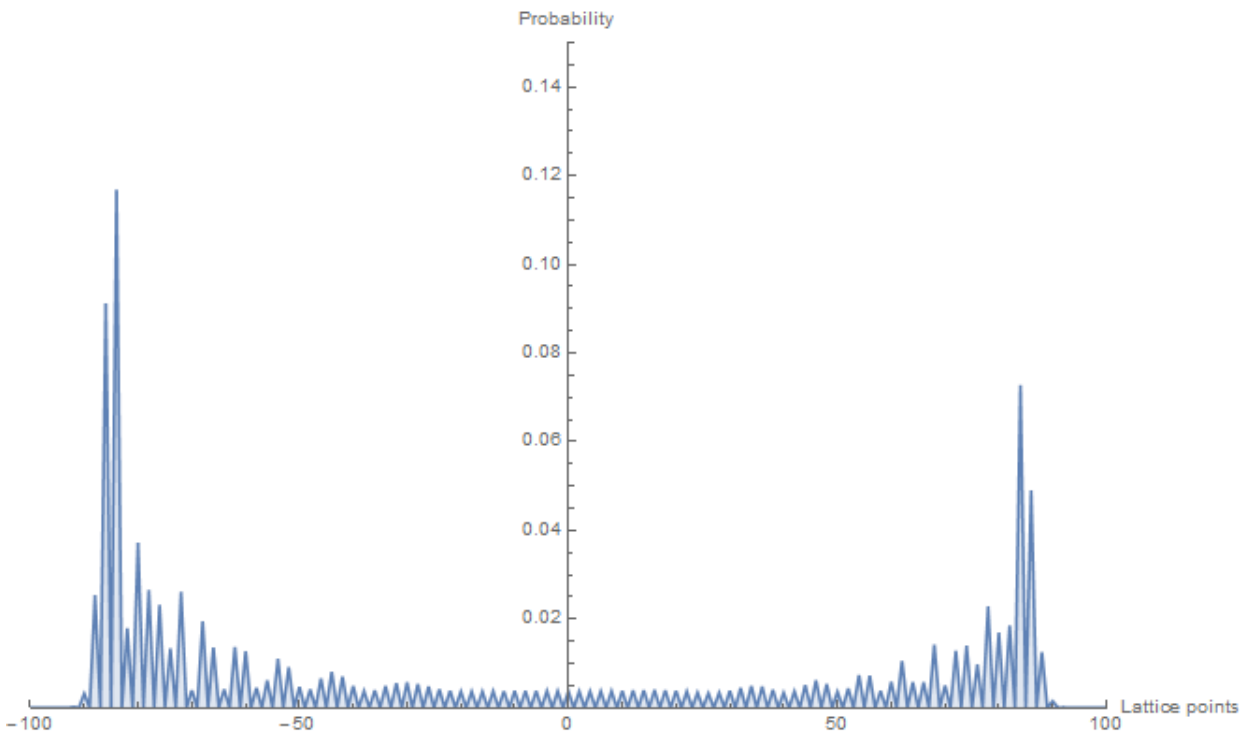

Figure 6.4: Using a general unitary, (30,30,0) and skewness to the left.

For (0,30,30)

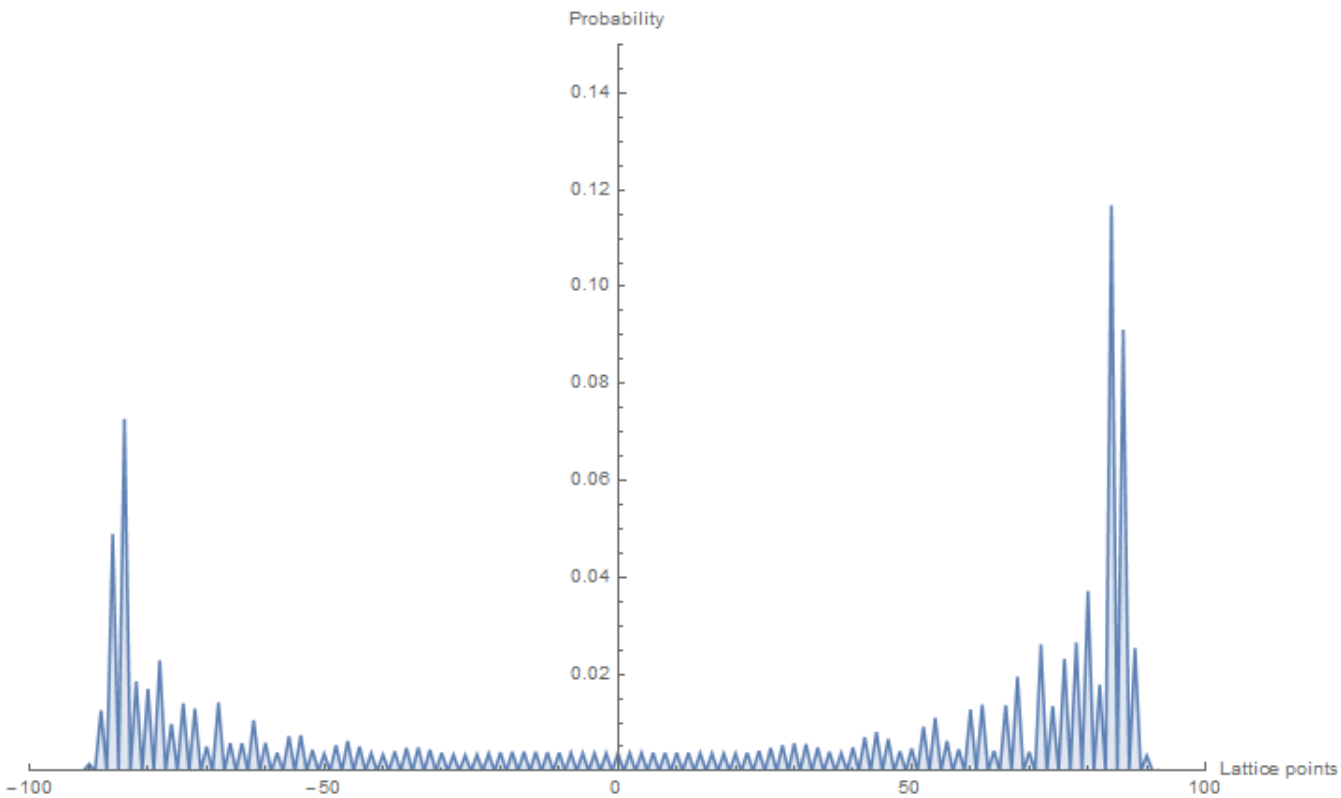

Figure 6.5: Using a general unitary, (0,30,30) and skewness to the left.

So one can control the direction of the skewness, either by carefully manipulating the initial state or the coin operator.

6.3 Quantum penny flip on a quantum random walk

Consider a one-dimensional quantum random walk on a line. The initial state of the coin is(say) $|0\rangle$ at position x=0 i.e.:

$$\psi = |0_c\rangle \otimes |0_x\rangle \tag{6.5}$$

The coin states are shared between the players A and B who make moves on the same coin. The player A is allowed to play quantum strategies on the coin that can put it in any superposition of $|0\rangle$ and $|1\rangle$ while the player B is allowed only classical strategies of flipping or not flipping the state of the coin.

Both the players share the same shift operator, S:

$$\hat{S} = |0\rangle_c\langle 0| \otimes \sum_i |i-1\rangle_p\langle i| + |1\rangle_c\langle 1| \otimes \sum_i |i+1\rangle_p\langle i| \tag{6.6}$$

which means if the spin is up or zero, the particle moves to the left and if the spin is down or one, the spin moves to the right.

The game is played in the following manner:

- Initialize the state of the system, i.e. a one-dimensional quantum random walker on a line with a single coin.
- Player A plays a quantum strategy say U on the coin and then applies the shift operator, changing the random walker's position.
- Player B then plays a classical strategy say X on the same coin and then applies the shift operator, changing the random walker's position again.
- Repeat the two steps above for N number of times.
- Measure the position of the particle and generate a probability distribution.
- If the distribution is asymmetrically biased towards the left then A wins, if it is asymmetrically biased towards the right then B wins else if it is symmetric then its a draw.

6.3.1 H vs I

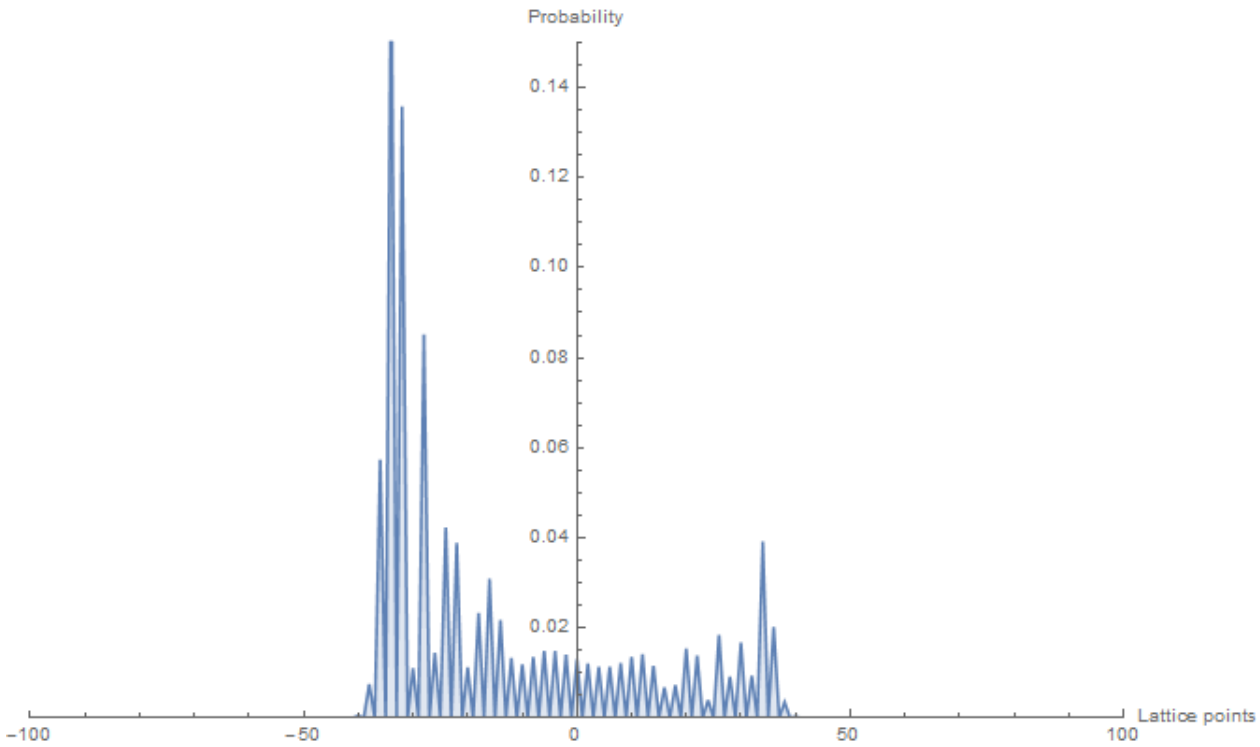

Figure 6.6: Player A plays Hadamard and Player B plays Identity one after the other for N = 100 on the initial state $|0\rangle$.

The distribution is the same as the original Hadamard walk, scaled to half its length.

6.3.2 H vs X

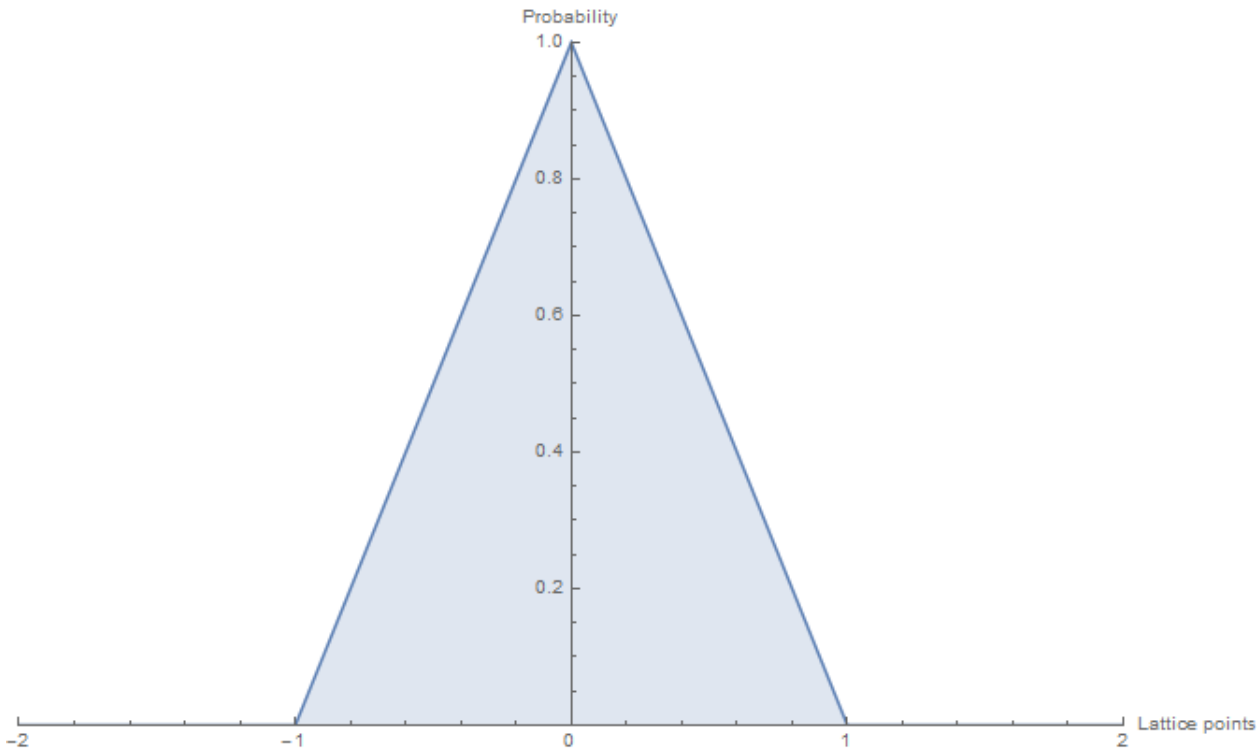

Figure 6.7: Player A plays Hadamard and Player B plays Flip one after the other for N = 100 on the initial state $|0\rangle$.

A unique complete localization of the particle at x = 0.

6.4 The entangled quantum walk as a game:

Consider a one-dimensional quantum random walk on a line. The initial state of the system is an entangled coin $\frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle)$ at position x=0 i.e.:

$$\psi = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle) \otimes |0_x\rangle \tag{6.7}$$

The entangled coin states are shared between the players A and B who make moves on their respective coins. The player A is allowed to play quantum strategies on the coin that can put it in any superposition of $|0\rangle$ and $|1\rangle$ while the player B is allowed only classical strategies of flipping or not flipping the state of the coin.

Both the players share the same shift operator, S:

$$\begin{aligned}\hat{S}_{EC} = |00\rangle_{cc}\langle 00| \otimes \sum_i |i+1\rangle_{pp}\langle i| + |01\rangle_{cc}\langle 01| \otimes \sum_i |i\rangle_{pp}\langle i| \\ +|10\rangle_{cc}\langle 10| \otimes \sum_i |i\rangle_{pp}\langle i| + |11\rangle_{cc}\langle 11| \otimes \sum_i |i-1\rangle_{pp}\langle i|\end{aligned} \tag{6.8}$$

The game is played in the following manner:

- Initialize the state of the system, i.e. a one-dimensional quantum random walker on a line with entangled coins.
- Player A plays a quantum strategy say U on his coin and then applies the shift operator, changing the random walker's position.
- Player B then plays a classical strategy say X on his coin and then applies the shift operator, changing the random walker's position again.
- Repeat the two steps above for N number of times.
- Measure the position of the particle and generate a probability distribution.
- If the distribution is asymmetrically biased towards the left then A wins, if it is asymmetrically biased towards the right then B wins else if it is symmetric then its a draw.

6.5 Introduction to entangled quantum random walks:

Quantum Hadamard walk on an entangled coin with the usual shift operator and initial state is $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

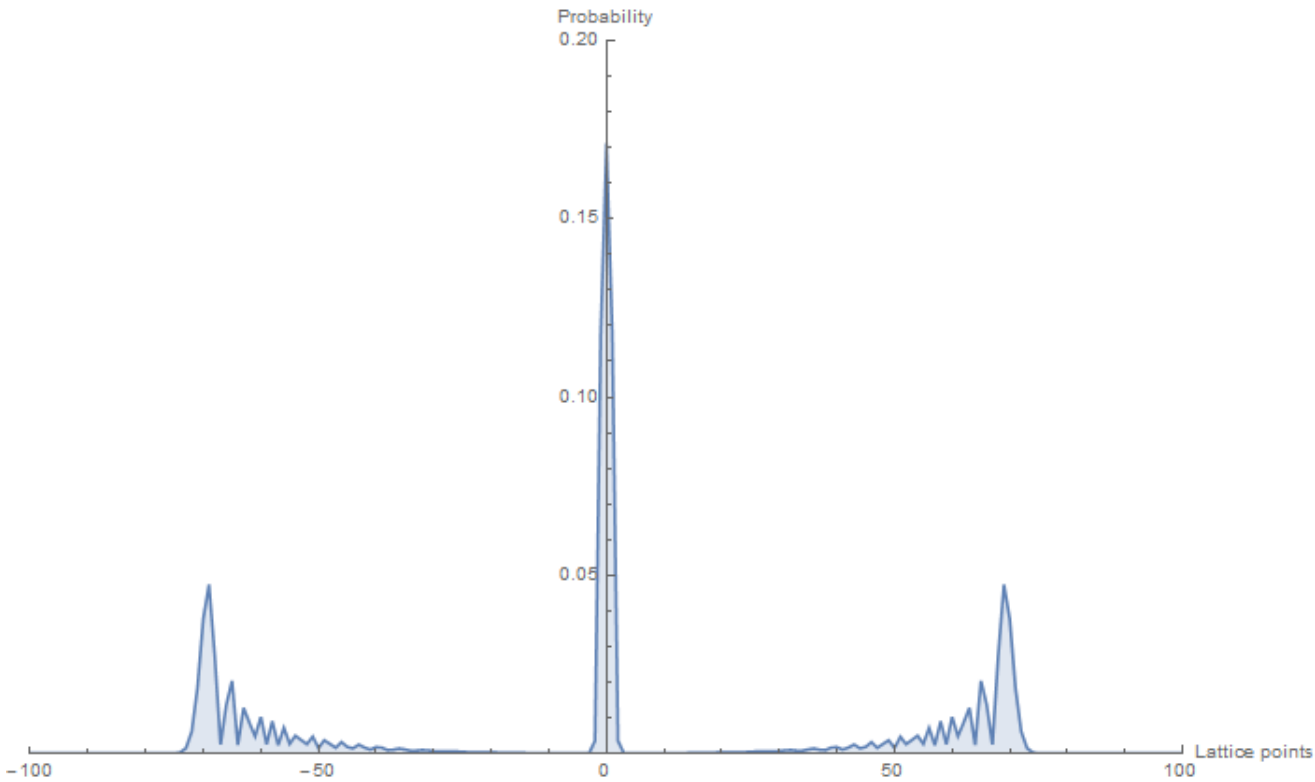

Figure 6.8: Player A plays Hadamard and so does Player B, one after the other for N = 100 on the initial state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

The three peak distribution.

6.5.1 Pure strategy: H vs X

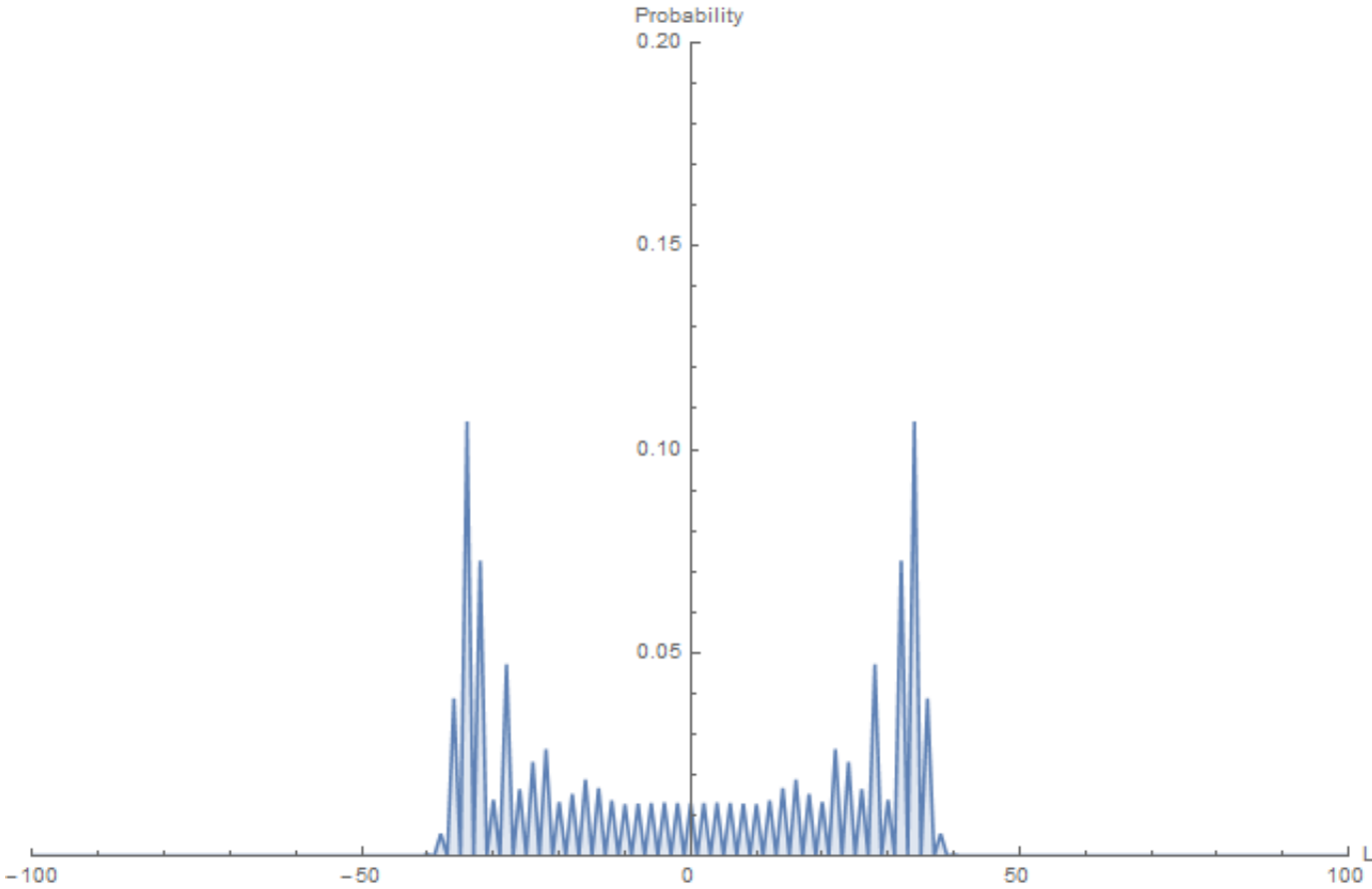

Figure 6.9: Player A plays Hadamard and Player B plays X, one after the other for N = 100 on the initial state$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

The particle has not gone beyond half the maximum distance it can travel i.e. 100 units. The peaks are localized near ± 40.

6.5.2 Pure strategy: H vs I

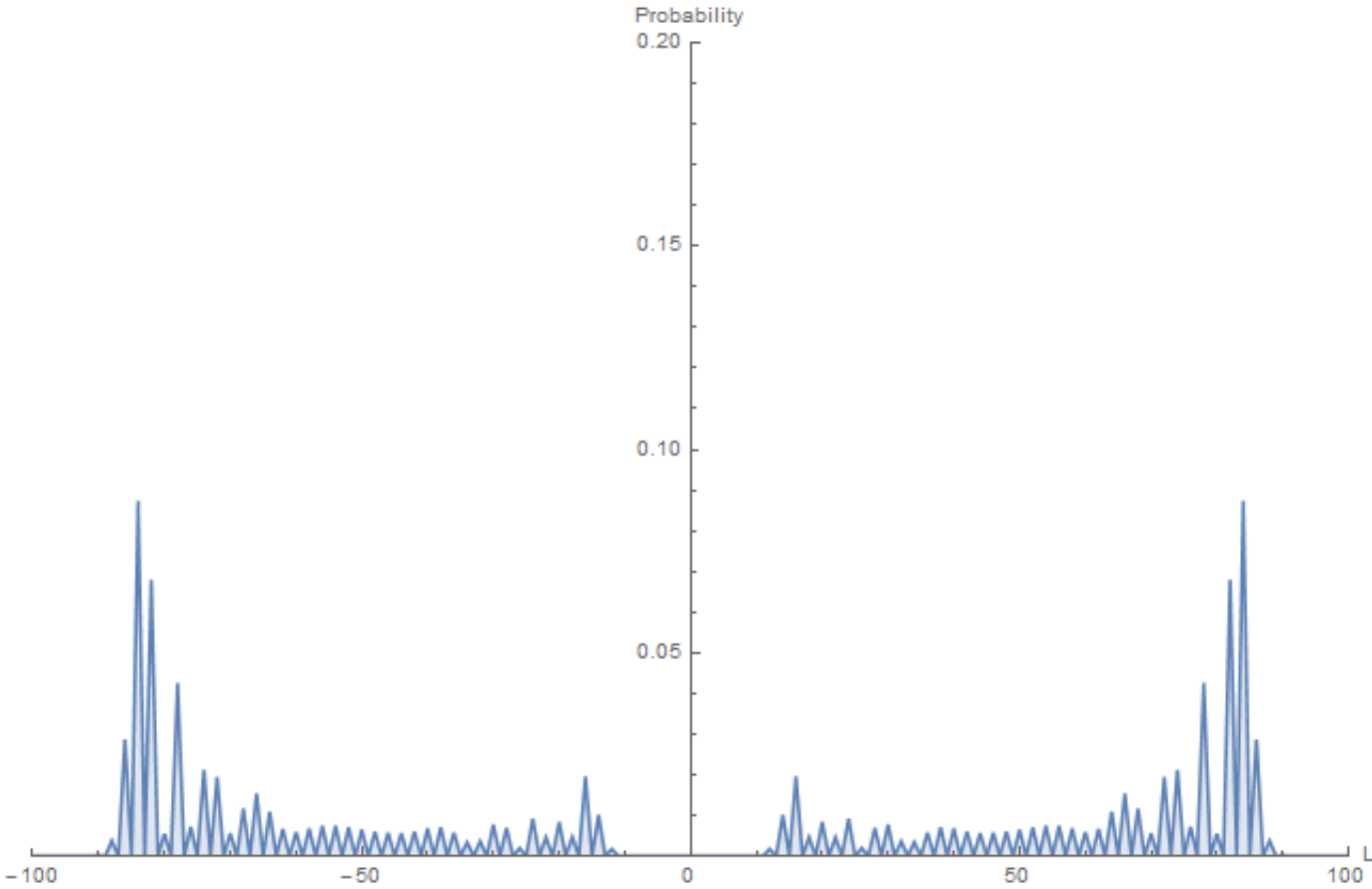

Figure 6.10: Player A plays Hadamard and Player B plays I, one after the other for N = 100 on the initial state$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

One can see that in contrast to the H vs I, now the peaks of the probability distribution are rather pushed towards the edges.

6.5.3 Mixed strategy: H vs (pI + (1-p)X)

One expects an average over the two strategies in this case.

mixed strategy p = 0.5

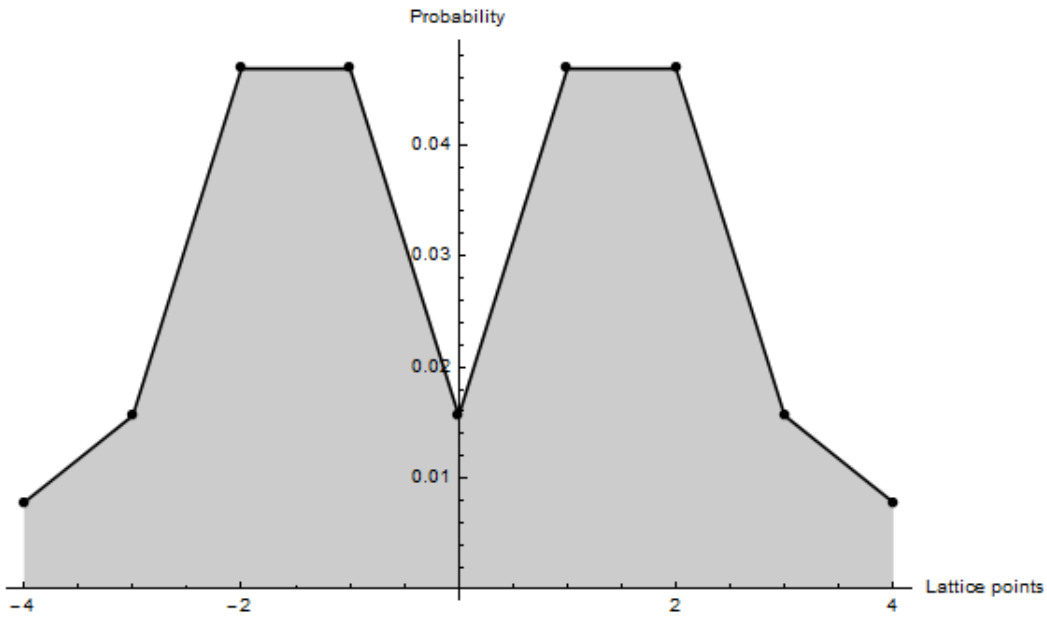

Figure 6.11: Player A plays Hadamard and Player B plays a mixed strategy of pI + (1-p)X for p = 0.5 , one after the other for N = 6 on the initial state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

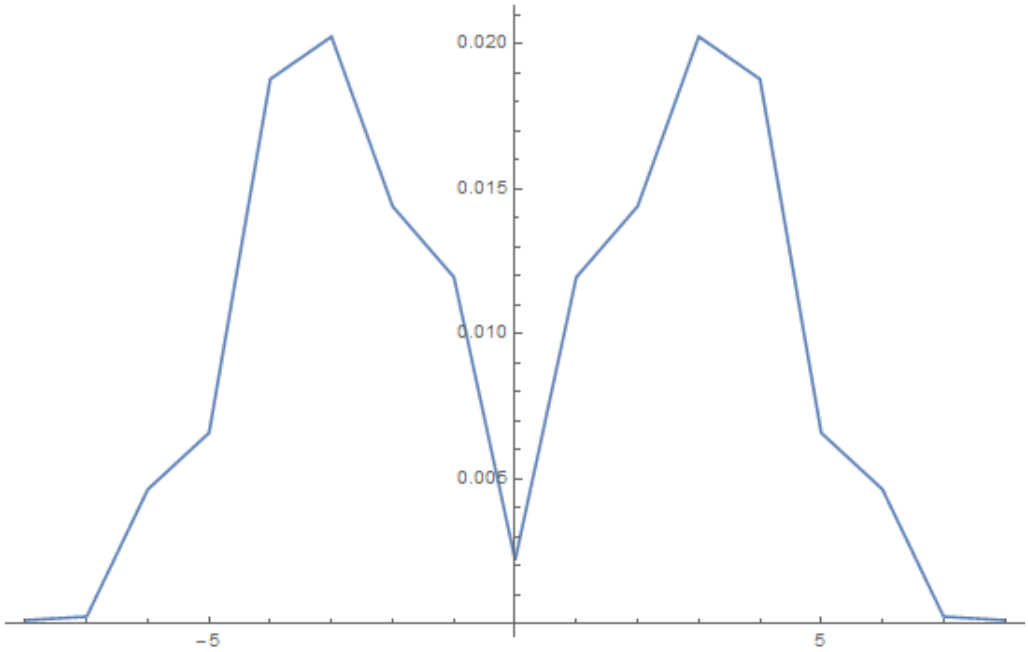

Figure 6.12: Player A plays Hadamard and Player B plays a mixed strategy of pI + (1-p)X for p = 0.5 , one after the other for N = 8 on the initial state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

The simulation was computationally very expensive and as a result, we could only go so far as N = 8. We expect that in the limit of N = 100, we would see the classical behavior.

6.6 Future plans

We wish to explore the mixed strategy quantum walk on entangled coins for N = 100.

References

[1] Quantum Notation, Version 2.3.0, A free Mathematica add-on for Dirac Bra-Ket Notation, Quantum Algebra, Quantum Computing and the QHD approximation to the Heisenberg Equations of Motion, available at http://homepage.cem.itesm.mx/lgomez/quantum/

Chapter 7

Conclusion

In conclusion, we have explored several ideas from Quantum game theory in the field of quantum walks. This gave us some new results like the complete localization of the particle at $x = 0$ in the H vs X case for a single coin operator. Similarly in the entangled penny flip game as a quantum walk, we expect a classical behaviour in the large N limit.

Appendix A

Mathematica codes

The codes for all the figures/simulations in the thesis are available upon request. Kindly send an email to either:
namit.anand@niser.ac.in or to colin@niser.ac.in

Figure 7.1: Quantum Hadamard walk with initial state $|0\rangle$

```
A = Array[a, {201, 2}, 0];
B = Array[b, {201, 2}, 0];
For[i = 0, i < 201, i++, For[j = 0, j < 2, j++, a[i, j] = 0]]
For[i = 0, i < 201, i++, For[j = 0, j < 2, j++, b[i, j] = 0]]
a[100, 0] = 1;

For[k = 0, k < 100, k++, For[i = 0, i < 201, i++, For[j = 0, j < 2, j++, b[i, j] = 0]];
 (*If[EvenQ[k],*)
 For[i = 0, i < 201, i++, If[(a[i, 0] ≠ 0 || a[i, 1] ≠ 0), (a0 = a[i, 0] ;
    a1 = a[i, 1];
    a[i, 0] = ((a0 + a1) / Sqrt[2]);
    a[i, 1] = ((a0 - a1) / Sqrt[2])), ]] ;
For[i = 0, i < 201, i++, If[a[i, 0] ≠ 0, b[i - 1, 0] = a[i, 0], ];
  If[a[i, 1] ≠ 0, b[i + 1, 1] = a[i, 1], ]];
 For[i = 0, i < 201, i++, For[j = 0, j < 2, j++, a[i, j] = b[i, j]]]]

(*Export["hadamard.csv",N[A]]*)
For[i = 0, i < 201, i++, For[j = 0, j < 2, j++, a[i, j] = a[i, j] * a[i, j]]]
For[i = 0, i < 201, i++, a[i, 0] = a[i, 0] + a[i, 1]]

(*Spin up*)
ListLinePlot[Table[{i - 100, N[a[i, 0]]}, {i, 0, 200}],
 PlotRange → {{-100, 100}, {0, 0.15}}, Filling → Axis,
 AxesLabel → {"Lattice points", "Probability"}]

(*Spin down*)
(*ListLinePlot[Table[{i-100,N[a[i,1]]},{i,0,200}],
 PlotRange→{{-100,100},{0,0.15}}]*)
```

Figure 7.2: Quantum entangled walk for H vs H

```
A = Array[a, {201, 4}, 0];
B = Array[b, {201, 4}, 0];
For[i = 0, i < 201, i++, For[j = 0, j < 4, j++, a[i, j] = 0]]
For[i = 0, i < 201, i++, For[j = 0, j < 4, j++, b[i, j] = 0]]
a[100, 0] = 1 / Sqrt[2];
a[100, 3] = 1 / Sqrt[2];
(*a[100,0]=1/2;
a[100,1]=1/2;
a[100,2]=1/2;
a[100,3]=1/2;*)


For[k = 0, k < 100, k++, For[i = 0, i < 201, i++, For[j = 0, j < 4, j++, b[i, j] = 0]];
  For[i = 0, i < 201, i++,
   (*If[a[i,0]≠ 0 || a[i,1]≠ 0||a[i,2]≠ 0||a[i,3]≠ 0,*)(a0 = a[i, 0] ;
    a1 = a[i, 1];
    a2 = a[i, 2];
    a3 = a[i, 3];
    a[i, 0] = 1 / 2 * (a0 + a1 + a2 + a3);
    a[i, 1] = 1 / 2 * (a0 - a1 + a2 - a3);
    a[i, 2] = 1 / 2 * (a0 + a1 - a2 - a3);
    a[i, 3] = 1 / 2 * (a0 - a1 - a2 + a3))(*,  ]*)];
  (*Export["hadamard.csv",N[A]]*)

  For[i = 0, i < 201, i++, If[a[i, 0] ≠ 0, b[i - 1, 0] = a[i, 0], ];
   If[a[i, 1] ≠ 0, b[i, 1] = a[i, 1], ];
   If[a[i, 2] ≠ 0, b[i, 2] = a[i, 2], ];
   If[a[i, 3] ≠ 0, b[i + 1, 3] = a[i, 3], ]];
  For[i = 0, i < 201, i++, For[j = 0, j < 4, j++, a[i, j] = b[i, j]]]];

For[i = 0, i < 201, i++, For[j = 0, j < 4, j++, a[i, j] = a[i, j] * a[i, j]]];

For[i = 0, i < 201, i++, a[i, 0] = a[i, 0] + a[i, 1] + a[i, 2] + a[i, 3]];
ListLinePlot[Table[{i - 100, N[a[i, 0]]}, {i, 0, 200}],
 PlotRange → {{-100, 100}, {0, 0.2}}, Filling → Axis,
 AxesLabel → {"Lattice points", "Probability"}]
(*ListLinePlot[Table[{i-100,N[a[i,1]]},{i,0,200}],
  PlotRange→{{-100,100},{0,0.1}}]
 ListLinePlot[Table[{i-100,N[a[i,2]]},{i,0,200}],PlotRange→{{-100,100},{0,0.1}}]
 ListLinePlot[Table[{i-100,N[a[i,3]]},{i,0,200}],
  PlotRange→{{-100,100},{0,0.1}}]*)
```